

Topical gel containing phenolic-rich extract from *Ipomoea pes-capre* leaf (Convolvulaceae) has anti-inflammatory, wound healing, and antiophidic properties

Jacinthia Beatriz Xavier-Santos^a, Júlia Gabriela Ramos Passos^a, Jacyra Antunes Santos Gomes^a, Joelly Vilaine Cavalcante Cruz^a, Jovelina Samara Ferreira Alves^b, Vinícius Barreto Garcia^c, Rodrigo Moreira da Silva^d, Norberto Peoporine Lopes^d, Raimundo Fernandes Araujo-Junior^c, Silvana Maria Zucolotto^b, Arnóbio Antônio Silva-Junior^a, Juliana Félix-Silva^a, Matheus Freitas Fernandes-Pedrosa^{a,*}

^a Laboratório de Tecnologia & Biotecnologia Farmacêutica (TecBioFar), Faculdade de Farmácia, Universidade Federal do Rio Grande do Norte (UFRN), Natal, RN, Brazil

^b Grupo de Pesquisa em Produtos Naturais Bioativos (PNBio), Laboratório de Farmacognosia, Faculdade de Farmácia, Universidade Federal do Rio Grande do Norte (UFRN), Natal, RN, Brazil

^c Departamento de Morfologia, Universidade Federal do Rio Grande do Norte (UFRN), Natal, RN, Brazil

^d Departamento de Ciências Biomoleculares, Faculdade de Ciências Farmacêuticas de Ribeirão Preto, Universidade de São Paulo, Ribeirão Preto, Brazil

ARTICLE INFO

Keywords:

Ipomoea pes-caprae
Herbal gel
Phytotherapeutic
Wound healing
Anti-inflammatory
Bothrops

ABSTRACT

The growing use of phytotherapy in clinical practice arouses interest in studies using medicinal plants as active ingredients for new medicines. *Ipomoea pes-caprae* has a wide medicinal use in the treatment of inflammatory disorders, skin wounds, stings, and painful rheumatic processes. Assayed in this study are the physicochemical characterization of a gel developed with this extract and the evaluation of its anti-inflammatory and healing efficacy, in addition to its antiedematogenic action on *Bothrops* snake envenoming in mice. The qualitative and quantitative analyses of the hydroethanolic extract by mass spectrometry showed 18 phenolic compounds, highlighting a high content of chlorogenic acid (0.92 µg/g), neochlorogenic acid (6.07 µg/g), and isochlorogenic acid (0.80 µg/g) compounds. The formulation was stable in relation to the physical-chemical characteristics during the time of analysis and was considered safe for topical treatment in animals, causing no skin irritation. Although the results have shown an absence of activity in the model of ear edema induced by croton oil (acute inflammation), the herbal gel efficiently inhibited carrageenan paw edema and chronic ear edema induced by multiple applications of croton oil, which may indicate the possible performance under the kinin pathway such as bradykinin, histamine, and serotonin. Wound healing in the group treated with the *I. pes-caprae* gel was accelerated compared with the placebo group, also confirmed through histological data. Edema induced by *Bothrops erythromelas* snake venom was efficiently reduced in the treatment with *I. pes-caprae* gel associated with the antithrombotic-crotalic serum, whereas the antivenom alone was not effective. This approach presents a promising formulation based on *I. pes-caprae* with potential therapeutic use for inflammatory disorders.

Abbreviations: ANOVA, analysis of variance; ANVISA, National Health Surveillance Agency; CGEN, Brazilian Genetic Heritage Management Council; CIOMS, Council of International Organizations of Medical Sciences; CONCEA, National Council for the Control of Animal Experimentation of Brazil; EDTA, ethylenediamine tetraacetic acid; HPLC, high-performance liquid chromatography; MDA, malondialdehyde; MPO, myeloperoxidase; PBS, phosphate buffered saline; SEM, standard error of mean; SISBIO, Brazilian Biodiversity Authorization and Information System; SRATs, substances reactive to tissue thiobarbituric acid.

* Correspondence to: Av. Gal. Cordeiro de Farias, s/n, Petrópolis, CEP 59012-570 Natal, RN, Brazil.

E-mail addresses: jacinthia.santos.110@ufrn.edu.br (J.B. Xavier-Santos), julia.passos.098@ufrn.edu.br (J.G.R. Passos), jacyra.gomes.015@ufrn.edu.br (J.A.S. Gomes), joelly.cruz.080@ufrn.net.edu.br (J.V.C. Cruz), jovelina.alves.054@ufrn.edu.br (J.S.F. Alves), vinicius.garcia.093@ufrn.edu.br (V.B. Garcia), rodeira@usp.br (R.M. da Silva), npelopes@fcfrp.usp.br (N.P. Lopes), fernandes.araujo@ufrn.br (R.F. Araujo-Junior), silvanazucolotto@ufrnet.br (S.M. Zucolotto), arnobiosilva@ufrnet.br (A.A. Silva-Junior), juliana.felix@ufrn.br (J. Félix-Silva), matheus.pedrosa@ufrn.br (M.F. Fernandes-Pedrosa).

<https://doi.org/10.1016/j.bioph.2022.112921>

Received 10 March 2022; Received in revised form 29 March 2022; Accepted 31 March 2022

Available online 7 April 2022

0753-3322/© 2022 Published by Elsevier Masson SAS. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Medicinal plants make an important contribution to health around the world. The use of natural products for the alternative treatment of diseases [1]. The knowledge about medicinal plants is often the only therapeutic resource available to many people [2,3], and according to the World Health Organization (WHO), the populations of developing countries make use of medicinal plants due to the difficulty of accessing prescription drugs. In rural areas or villages, using medicinal plants has become, in many cases, the only form of access to primary health care available [4].

Brazil, presenting an enormous biodiversity and covering about 20–22% of the entire planet, is associated with the vast traditional knowledge regarding the use of plants and it has great potential in the development of new herbal medicines for the market [5,6]. Several important synthetic medicines were discovered from plants, and in recent years, there has been an increasing acceptance of herbal medicines [7]. The use has been associated with its advantages of high therapeutic efficacy, low side effects, and, in some cases, lower cost compared to synthetic products [8]. This fact leads many researchers to investigate the potential of using plant species for the development of new products for the market.

Among the various plant species, *Ipomoea pes-caprae* (L.) R. Br. has wide popular medicinal use and stands out as a source of phenolic compounds. Belonging to the genus *Ipomoea* and the family Convolvulaceae, which in turn has about 500–600 species, it is popularly known “morning glory” and in Brazil as “salsa-da-praia” or “pé-de-cabra” [9,10]. *I. pes-caprae* is a small plant, native to Brazil and widely distributed in the coastal dune region. Previous studies have demonstrated its use for various purposes in traditional medicine, especially in the treatment of itching, skin wounds, painful and inflammatory processes, stings, burns caused by jellyfish, and rheumatisms [9–15]. Biological activities are generally related to the presence of flavonoids, alkaloids, carbohydrates, tannins, sterols, and terpenoids [16–18].

In this study, *I. pes-caprae* leaf extract was incorporated into a gel developed for topical application, the physical-chemical characterization was performed, and the pharmacological activity was evaluated in *in vivo* models of inflammation, wound healing, and edema induced by snake venom, with the objective of demonstrating several therapeutic applications of an herbal formulation, based on its popular use and previous studies in the literature.

2. Materials and methods

2.1. Chemicals and reagents

Croton oil, dimethylsulfoxide, hexadecyltrimethylammonium bromide, hydroxypropylmethylcellulose, naphthyl ethylenediamine dihydrochloride, *o*-dianisidine, poloxamer F-127, sodium nitrite, sulfanilamide, thiobarbituric acid, and trichloroacetic acid were purchased from Sigma-Aldrich (St. Louis, MO, USA). Hydrogen peroxide was purchased from Merck (Darmstadt, HE, Germany). Dexamethasone was purchased from Aché (Guarulhos, SP, Brazil), ketamine from Vetnil (Louveira, SP, Brazil), and xylazine from Agener União Saúde Animal (São Paulo, SP, Brazil). All other reagents and solvents used were of analytical grade. The water used was purified by reverse osmosis.

2.2. Animals

Swiss albino mice (30–35 g, 6–8 weeks old) of both sexes, supplied by the animal facility of the Health Sciences Center from the Federal University of Rio Grande Norte, were used. Animals were housed in standard polypropylene cages and maintained under controlled temperature (22 ± 2 °C) in a 12 h light/ dark cycle. Mice were fed with standard laboratory extruded food and water *ad libitum*. At the end of the experiments, the animals were euthanized by intraperitoneal anesthetic

overdose (xylazine 150 mg/kg plus ketamine 300 mg/kg).

All procedures involving animals were approved by the Ethics Committee on Animal Use from the Federal University of Rio Grande Norte (protocol no. 091.072/2017) and performed in agreement with the recommendations of the Brazilian National Council for the Control of Animal Experimentation (CONCEA) and the International Guiding Principles for Biomedical Research Involving Animals of the Council of International Organizations of Medical Sciences (CIOMS).

2.3. Plant material

2.3.1. Collection and botanical authentication

Ipomoea pes-caprae (L.) R. Br. leaves were collected in May 2017 in Natal city, Rio Grande do Norte state, Brazil (5°45'54"S, 35°11'46.63"W). The collection was authorized by the Brazilian Authorization and Biodiversity Information System (SISBIO) (process no. 35017) and the research was authorized by the National System for the Management of Genetic Heritage and Associated Traditional Knowledge (SISGEN) (process no. A4E8DE0). Botanical identification of the material was performed at the Herbarium in the Biosciences Center of the Federal University of Rio Grande Norte, where an authentic sample of *I. pes-caprae* was deposited under voucher number UFRN 431. Plant name was checked in <http://www.theplantlist.org> (accessed in August 2021). The leaves were dried at 45 °C in a circulating air oven one week, triturated and stored in hermetically sealed bottles protected from light and humidity until use for extract preparation.

2.3.2. Extract preparation

The extract of dried *I. pes-caprae* leaf was prepared by the maceration method, using a 70% hydroethanolic solution in a 1:10 ratio (drug: solvent, w/v) for 48 h at room temperature, and a remaceration by the same procedure. After this period, part of the extracts was filtered and used to perform the phytochemical study, and the other part had this volume reduced in rotary evaporated; then it was freeze dried and stored at –20 °C. The extraction method and the solvent were chosen according to the previous literature to reach a good yield and flavonoid extraction [19].

2.4. Phytochemical analysis of *Ipomoea pes-caprae* leaf extract

2.4.1. Qualitative analysis

The chemical profile of *I. pes-caprae* leaf extract was analyzed using a Shimadzu™ (Kyoto, Japan) High Performance Liquid Chromatography (HPLC) coupled to an amaZonSL ion trap (IT) Bruker Daltonics™ (Billerica, USA). The HPLC consists of a LC-20 CE solvent pump unit, a DGU-20A3 online degasser, a CTO- 20A column oven, a DGU-20A3 online degasser, a CBM-20A system controller, and an SPD-M20A (200–400 nm) diode array detector. Injections were carried out automatically (20 µL) through a 100 µL loop SIL-20A HT and the sample was solubilized in methanol: water (1:1; v/v) at concentration of 2 mg/mL and filtered through a polyvinylidene fluoride (PVDF) syringe filter. Compounds were separated at a temperature of 30 °C using a Luna C18 column (5 µm, 250 mm × 4.60 mm; Phenomenex™). The mobile phase used was A: Water and B: Acetonitrile both with added Acetic Acid (2%; v/v). The flow rate used was 1 mL/min, in a gradient elution: 5.0% B, 0–5 min and 5–100% B, 5–90 min.

2.4.2. Quantitative analysis

Quantitative analysis was carried out on a Prominence Liquid Chromatography system (Shimadzu, Kyoto, Japan) equipped with an online degasser DGU-20A3R, a solvent pump LC-AD, an autosampler SIL-20A8HT, a column oven CTO-20AC, and a controller CBM-20A. The chromatographic separation was achieved on an Ascentis Express C18 column 100 mm × 4.6 mm, 2.7 µm (Supelco, Bellefonte, EUA) and an equivalent guard column (5 mm, Phenomenex, United States). Gradient elution with 0.1% formic acid (A) and methanol (B) was applied for

separation of compounds at a flow rate of 1.0 mL/min. The mobile phase used was 15.0% B from 0 to 3.0 min and 15–30% B from 3.0 to 10 min, and the injection volume was 5 μ L. The extract (20 μ g/mL) and neochlorogenic acid (5-O-caffeoylquinic acid; 2–300 ng/mL) were diluted in methanol: water (1:1; v/v) and filtered through a polytetrafluoroethylene membrane with 0.22 μ m of pore size before the chromatography analysis. Carryover from subsequent injections was not observed. An Applied Biosystems/Sciex API 3200 triple quadrupole Mass Spectrometer (AB Sciex, Concord, Canada) (LC-MS/MS) was used in this study. The MS was operated in the negative ion and multiple reaction monitoring (MRM) mode for the transition 353.137 $\rightarrow$ 191.000 for quantitative analysis of neochlorogenic, chlorogenic, and isochlorogenic isomeric acids (m/z 354.31). The method was validated in terms of linearity, specificity, precision, accuracy, limit of detection (LOD), limit of quantification (LOQ), and repeatability according to the considerations of the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH guidelines) [20–22].

2.5. *In vitro* cytotoxicity assay

2.5.1. Hemolytic assay

This assay was performed as previously described in the literature [23]. First, a human erythrocyte suspension was prepared from human blood collected in 2 mg/mL K3-EDTA tubes (VACUETTE® - São Paulo, SP, Brazil) and centrifuged at 1300 g for 10 min. Plasma obtained and the top layer of cells were discarded. The erythrocytes were resuspended in 10% (v/v) phosphate buffered saline (PBS), homogenized, and centrifuged again at 2.500 g for 10 min. This process was repeated until the supernatant became clear. Finally, the erythrocytes were resuspended in 10% (v/v) PBS. For the hemolytic assay, the hydroethanolic extract of *I. pes-caprae* in different concentrations (62.5, 125, 250, 500, and 1000 μ g/mL) were placed in microtubes together with 20 μ L of the erythrocyte suspension. For control of 100% hemolysis, microtubes were prepared with the erythrocyte suspension and purified water. For negative control, microtubes with erythrocyte suspension and PBS were used. After this process, the microtubes were incubated at 37 °C for 60 min. These samples were centrifuged at 1.300 g for 3 min at room temperature. And finally, 100 μ L of the supernatant of these samples were read in a microplate reader at 540 nm. Human blood collection from healthy donors were performed upon approbation from the Research Ethics Committee of the Federal University of Rio Grande do Norte (CAAE 88074318.3000.5292).

2.5.2. Cell viability

Cell viability over the cell line of fibroblasts (NIH/3T3, ATCC CRL-1658) was evaluated by the 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl-tetrazolium (MTT) assay [24]. The cells were cultured at 37 °C in a 5.0% CO₂ atmosphere until confluence in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum. In a 96-well plate, 100 μ L of the cell suspension (7×10^3 cells/well) in DMEM medium supplemented with 10% fetal bovine serum was incubated under the conditions described above for 24 h. Then, the medium was replaced by 100 μ L of DMEM in the absence of bovine serum for 24 h at 37 °C in an atmosphere at 5.0% CO₂. After incubation, the medium was removed and 100 μ L of DMEM medium supplemented with 10% fetal bovine serum (viability control) or extract of *I. pes-caprae* (250–0.031 mg/mL) was added. The mixture was maintained under the conditions described above for 24 h. Then, the content of the wells was replaced with 100 μ L of the MTT solution (1 mg/mL) and incubated at 37 °C for 4 h in a 5.0% CO₂ atmosphere. The formazan crystals were solubilized with 100 μ L of 96% ethyl alcohol (v/v) and kept at room temperature for 15 min, under stirring (40 rpm). Absorbance was measured at 570 nm in a microplate reader (Epoch Biotek, Winooski, USA). The percentage of cell viability was determined using the following formula: % Cell viability = Absorbance of sample

cells/Absorbance of viability control cells $\times$ 100.

2.6. Development of formulations

The topical gel formulation was prepared as previously described in the literature [25]. The dried leaf extract of *I. pes-caprae* was incorporated at 5.0% (w/w) (Ipc gel). The formulation used was: hydroxypropylmethylcellulose (0.2%, w/w), propylene glycol (0.3%, w/w), poloxamer F-127 (1.5%, w/w), isopropyl alcohol (0.5%, w/w), and methylparaben (0.02%, w/w). A placebo gel (without extract) was used as control. After preparation, the formulations were maintained at 8 °C, protected from light and humidity, for at least 24 h before physicochemical analysis or *in vivo* assays.

2.7. Physicochemical characterization of formulations

The formulations were subjected to analysis of organoleptic characteristics, centrifugation test, and pH determination. The developed formulations were evaluated for appearance, color, and odor after 24 h of their preparation for 30 days. The centrifugation test was performed in a microcentrifuge (Eppendorf, 5415 R, Hamburg, Germany), using 1 g of sample at a rotation of 800 g for 30 min, at room temperature. After centrifugation, the occurrence of changes in the visual aspect of the formulation was observed. The pH was determined using a 10% (w/v) dilution of each formulation in water in a pH meter (PHS-3E, Ionlab, Araucária, PR, Brazil), at room temperature, in triplicate, evaluating for 7 days.

2.8. Skin irritation test of formulations

The skin irritation test of developed formulations was conducted as previously described in the literature [25]. Mice were anesthetized with ketamine (90 mg/kg) and xylazine (10 mg/kg) intraperitoneally, and the dorsal back was shaved without causing tissue damage. Then, the dorsum of the mice was washed with 1% detergent solution and after dried, a test area of approximately 3 cm² was delimited. Filter papers (2 cm²) were impregnated with 100 mg of placebo or Ipc gel and then positioned on the dorsum of the animals, secured with hypoallergenic tape. Samples were maintained in contact with animals' skin for 4 h and then removed with 1% detergent solution. After 1, 24, 48, and 72 h, all animals were visually examined for signs of erythema, edema, dry skin, and any local or systemic clinical adverse effects, including behavioral changes, body weight, food, and water intake. After this period, the monitoring was performed on alternate days, until the 14th day. Purified water was used as negative control, and each group consisted of 3 animals.

2.9. Evaluation of anti-inflammatory activity

2.9.1. Croton oil single application-induced mouse ear edema (acute inflammation)

The test was performed according to the method previously described in the literature [25]. The animals received 20 μ L of a fresh 2.5% croton oil solution, applied topically to the inner surface of the right ear, while in the left ear an equal volume of acetone (vehicle) was applied as a control. The treatment consisted of applying 20 mg of the placebo gel, Ipc gel, or 1 mg/g dexamethasone cream (commercially purchased reference anti-inflammatory drug) in the animals' ears 15 s after edema induction. All applications were performed with a swab, using standardized movements of 25 frictions. Topical treatment was reapplied every 2 h after induction of edema. Edema was expressed as the percentage difference between the weight of the inflamed right ear and the non-inflamed left ear of each mouse. Then, mouse ears were stored at -80 °C.

Frozen ears were chopped and homogenated in 500 μ L of PBS. Then, samples were centrifuged at 10,000 g at 4 °C for 10 min. The pellets were

further employed for myeloperoxidase (MPO) enzyme extraction and quantification, as an indirect marker of neutrophil migration as described in the literature with a few modifications [24,25]. Initially, the pellets were homogenized in 500 μL of 0.5% hexadecyltrimethylammonium bromide in 50 mM potassium phosphate pH 6.0; afterwards, they were sonicated in ice bath for 30 s and submitted to 3 freeze-thaw cycles. At the end of the last freeze-thaw cycle, the samples were sonicated once more for 30 s in ice bath. Centrifugation was performed at 10,000 g for 10 min at 4 °C and the supernatant obtained was used for MPO activity determination, by mixing 20 μL of each sample with 200 μL of 50 mM potassium phosphate pH 6.0 containing 0.0005% hydrogen peroxide and 0.167 mg/mL o-dianisidine. MPO activity was colorimetrically determined using a microplate reader (Epoch-Biotek, Winooski, VT, USA), through kinetic reading at 1 min intervals, during 5 min. One unit of MPO activity was defined as the equivalent to the consumption of 1 μmol of hydrogen peroxide per minute, considering that 1 μmol of hydrogen peroxide gives a change in absorbance of 1.13×10^{-2} per minute [26]. The results were expressed by the units of MPO activity per ear.

2.9.2. Carrageenan-induced paw edema (acute inflammation)

Mice received a subplantar injection of 50 μL of a 1% (w/v) suspension of λ -carrageenan in the right hind paw [26]. After 30 s, the mice were treated topically in the paw where inflammation was induced with 20 mg of the placebo gel, Ipc gel, or 1 mg/g dexamethasone (anti-inflammatory positive control) by means of standardized movements of 25 frictions. Topical treatment was reapplied every 2 h after induction of edema. Paw thickness was measured by a digital caliper (Digimess, São Paulo, SP, Brazil) immediately prior to carrageenan injection and then at 1, 2, and 4 h after that. Edema was expressed as the percentage difference between the paw thickness after (at respective times) and before (baseline) carrageenan injection. At the end of the experiment, the animals were euthanized, and their paws were collected and stored at –80 °C, until processed for quantification of MPO activity, as described above in Section 2.9.1.

2.9.3. Croton oil multiple application-induced mouse ear edema (chronic inflammation)

This study was conducted in 9 days. Croton oil 5.0% (v/v) in acetone (20 μL /ear) was applied in the right ears and acetone in the left ears of Swiss mice with a micropipette on alternate days. The animals were treated with 20 mg of topical formulation of placebo gel, Ipc gel, or 1 mg/g dexamethasone (anti-inflammatory reference) for 4 days, 2 times a day, from the 5th day of the experiment until the last one. On day 9, the mice were euthanized, and 6 mm diameter ear punch biopsies were collected and weighed [27]. For edema quantification, ear punch biopsy was individually weighed using an analytic balance. Edema was expressed as the percentage difference between the weight of the inflamed right ear and the non-inflamed left ear of each mouse. Also, part of the samples was fixed in formaldehyde solution for histopathological analysis.

Frozen ears were chopped and homogenized in 500 μL of PBS. Then, samples were centrifuged at 10,000 g at 4 °C for 10 min. The supernatants were used for quantification of thiobarbituric acid reactive substances (TBARS), as a marker of oxidative damage to lipids (lipid peroxidation), as previously described in the literature with a few modifications [28]. Ear homogenates (100 μL) were mixed with 200 μL of a solution containing 15% trichloroacetic acid, 0.375% thiobarbituric acid, and 0.25 N hydrochloric acid for 30 s. After this, the samples were heated at 100 °C for 60 min, and the reaction was stopped by incubation in ice bath for 20 min. Centrifugation was performed at 10,000 g for 10 min at 4 °C and the supernatant obtained was read at 535 nm in a microplate reader for determination of malondialdehyde (MDA) equivalents, using its molar extinction coefficient at this wavelength, which is equal to $15,600 \text{ M}^{-1}\text{cm}^{-1}$. The results were expressed as nmol of MDA equivalents per ear.

2.9.3.1. Histopathological analysis. Biopsies were fixed in 10% buffered formaldehyde solution, dehydrated, and embedded in paraffin. Histological sections were stained with hematoxylin-eosin staining and examined by light microscopy (magnification 100 and 400x, Olympus BX50, attached to a Nikon DS-Fi2 camera) to assess the intensity of edema and leukocyte infiltration. Using ImageJ 1.44p software (Health Institutes, MD, USA) in the quantitative analysis, 10 representative areas of each ear biopsy were analyzed at 40x magnification to measure the distance between the epithelium of the inner and outer face of the ears, in micrometers, as a quantitative index of edema.

2.10. Wound-healing activity

The wound healing potential of the Ipc gel was evaluated, as already described in the literature with some changes [31]. Each mouse was anesthetized with the mixture of xylazine (10 mg/kg) and ketamine (80 mg/kg) by intraperitoneal route. Then, the shaved region was wiped with a sufficient amount of 70% ethanol. A circular fragment of dorsal skin with a diameter of 6 mm was removed with a punch. After the incision was made, the animals were treated topically twice a day for 8 days, and the dorsal area was photographed for later analysis on days 1, 3, 5 and 8. Topical treatment was made with 20 mg of the placebo gel or Ipc gel. The animals were euthanized on the eighth day. After euthanasia, an elliptical excision covering the area of the healing process was removed from each animal. The measurement of the wound area was taken based on the digital photographs of the mouse's wounds using the image analysis software ImageJ 1.44p (National Institutes of Health, MD, EUA). Also, part of the samples was fixed in formaldehyde solution for histopathological analysis.

2.11. Inhibition of edematogenic activity of *Bothrops erythromelas* snake venom

2.11.1. Venoms and antivenom

Venom from *Bothrops erythromelas* snake was kindly provided by Karla Patrícia de Oliveira da Luna (State University of Paraíba). Freeze-dried venom was stored at –20 °C until use. For the experiments, the venom was resuspended in PBS. The amount of venom was expressed by protein content, determined by Bradford protein assay using albumin as standard [29]. The scientific use of the venoms was approved by the National System for the Management of Genetic Heritage and Associated Traditional Knowledge (SISGEN) (process no. A4E8DE0). The commercial bothropic-crotalic antivenom was purchased from Lema Injex Biologic (Vespasiano, MG, Brazil). According to the package insert, this product is a solution purified by peptic and lyophilized digestion of specific immunoglobulins obtained from the serum of hyper-immunized horses with venoms from the snakes *B. jararaca* and *Crotalus durissus terrificus*, plus 0.5% of phenol as a preservative. The batch used was 013/15B.

2.11.2. Paw edema induction

The edematogenic activity of *B. erythromelas* snake venom was evaluated using the paw edema model, as previously described in the literature with a few modifications [30]. *B. erythromelas* (1.5 μg /50 μL of PBS) venom was injected subcutaneously into the right hind paw of mice. Control animals received equal volume of PBS (normal control). Venom concentration was chosen based on pilot assays, where the selected dose induced significant edema without producing hemorrhage. The individual right hind paw thickness was measured immediately before venom injection (basal value) and at selected time intervals after edema induction (0, 1, and 2 h), using a digital caliper (Digimess, São Paulo, SP, Brazil). The mice were treated topically in the paw where inflammation was induced, immediately after venom injection, with 20 mg of the placebo gel, Ipc gel, bothropic-crotalic antivenom (AV), or association AV + Ipc gel by means of standardized movements of 25 frictions. When used, AV (100 μL) was applied intraperitoneally. Edema

was expressed as the percentage difference between the thickness of the paw after (at respective time points) and before (basal values) venom injection, as mean $\pm$ standard error of mean, with $n = 5$ animals per group.

2.12. Statistical analysis

All results are presented as mean $\pm$ standard error of mean (SEM). One-way ANOVA followed by Tukey's test was performed using GraphPad Prism version 5.00 (San Diego, CA, USA), where p -values less than 0.05 were considered significant.

3. Results

3.1. Phytochemical analysis

The analysis of the phytochemical profile of *I. pes-caprae* extract showed 18 compounds (Fig. 1) that have been identified based on the MassBank database (<http://www.massbank.jp>) and comparison with literature data [31–40]. The MS analysis was accomplished in negative mode, and the spectral features of the chromatogram peaks are assembled in Table 2. The major compounds were identified as phenolic acids derived from benzoic and cinnamic acids (Table 1).

From the phytochemical profile of *I. pes-caprae* leaf extract, it is possible to suggest that phenolics acids is the major class of compounds present in extract. For this reason, the aim was to achieve quantifying the content of neochlorogenic, chlorogenic, and isochlorogenic acids (Fig. 2) in view of its potential application in pharmaceuticals related to human health.

The developed method to quantify the phenolic derivatives from *Ipomea pes-caprae* leaf extract was validated according to the instructions of the ICH guidelines [20–22]. The linear regression equations showed a fine linearity ($r > 0.99$) and concentration range for each reference standard, according to Table 2. The limit of quantification (LOQ) was defined as the ratio of 10:1, and the limit of detection (LOD) was determined as a signal-to-noise ratio of 3:1. The repeatability or intra-assay precision and the intermediate precision were evaluated by a set of sample (9 solutions) for the reference standard on 2 different days and by 2 technicians. Concentration data displayed relative standard deviations (RSD%) lower than 5% (Table 3). The repeatability and intermediate precision data were in agreement with the regulatory agency requirements. The accuracy was measured for 3 independent standard

Table 1

Phytochemical profile of *Ipomea pes-caprae* leaf extract characterized by HPLC-UV-DAD-MS/MS analysis.

No.	Rt (min)	[M-H] [−] m/z	MS ₂ fragments ions	$\gamma_{\max}$ (nm)	Identification
1	4.2	190.81	190	230	quinic acid
2	15.0	314.94	315, 233, 153	232	protocatechuic acid glucoside
3	19.6	352.93	191, 179, 135	233, 325	3-O-caffeoylquinic acid (chlorogenic)
4	23.9	341.05	179, 135, 233	233	caffeoyl-O-hexoside
5	28.5	353.02	191	233, 325	5-O-caffeoylquinic acid (neochlorogenic)
6	29.2	353.04	173, 179, 135	233, 325	4-O-caffeoylquinic acid (isochlorogenic)
7	30.4	178.87	178.87	233, 322	caffeic acid
8	32.2	499.14	343, 256, 173, 280	233	3-O-p-coumaroyl-4-O-caffeoylquinic acid
9	33.9	445.23	385.03	233, 330	putative of acacetin hexoside
10	36.4	367.10	191, 173	233	3-O-feruloyl quinic acid
11	37.1	385.09	267, 245	233	sinapoyl hexoside
12	41.0	304.97	304	233	epigallocatechin
13	43.9	515.08	353, 335	233, 327	4,5-di-O-caffeoylquinic acid
14	45.6	463.60	463, 301	233, 351	quercetin-3-O-glucoside
15	47.6	515.10	353, 399, 335, 256, 173	233, 328	3,5-di-O-caffeoylquinic acid
16	49.2	505.00	301, 463, 343, 179	233, 350	quercetin derivative
17	53.6	206.90	206	233, 324	3,4-dimethoxycinnamic acid
18	55.4	288.96	289	233	catechin or epicatechin

reference solutions prepared in triplicate and a value within the range of $\pm 10\%$ of the relative standard error (RSE%) was considered satisfactory, as shown in Table 3.

The quantitative analysis of phenolic compounds identified in *Ipomea pes-caprae* leaf extract showed high content of neochlorogenic, chlorogenic, and isochlorogenic acids compounds, according to Table 4. The developed LC-QqQ-MS method proved to be reliable for the analysis of phenolic compounds in *Ipomea pes-caprae* leaf extract. The parameters evaluated by this method such as linearity, precision, and accuracy gave satisfactory results, allowing its use for the quality control of raw

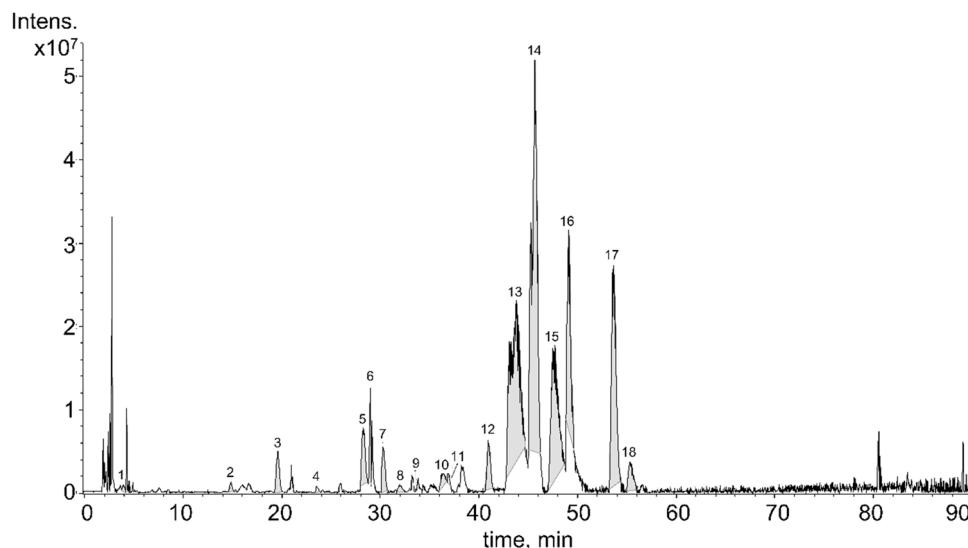


Fig. 1. LC-UV-ESI-IT-MS/MS total ion chromatogram of *Ipomea pes-caprae* leaf extract. Column: Luna C18 column (5 μ m, 250 mm $\times$ 4.60 mm; Phenomenex TM); mobile phase: A (2% CH₃COOH) and B (MeCN; 2% CH₃COOH), 5.0% B (0–5 min), 5–100% B (5–90 min).

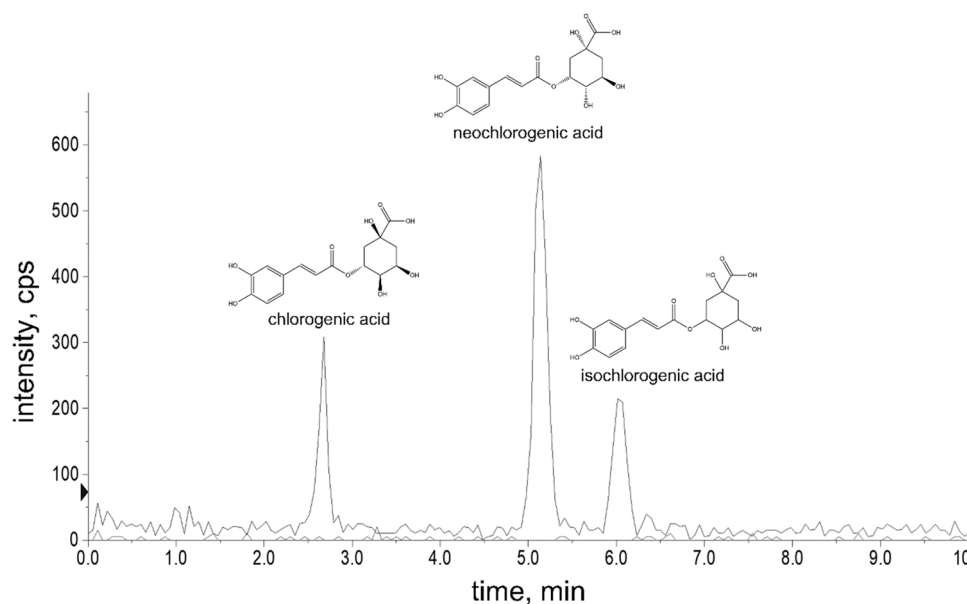


Fig. 2. LC-QqQ-MS (negative mode) chromatogram of *I. pes-caprae* leaf extract and product ion mass spectra of chlorogenic, neochlorogenic, and isochlorogenic acids.

Table 2

Linearity and sensitivity data.

Peak no.	Compound	Rt (min)	MS	Concentration Range ^a (ng/mL)	Regression Line	(r)	LOD (ng/mL)	LOQ (ng/mL)
1	neochlorogenic acid	5.10	[M-H] ⁻ m/z 353	2–300	$y = 102.11x + 96.3$	0.9982	0.75	2.5

^a Six levels of concentrations ($n = 3$); r , correlation coefficient; LOD, limit of detection; LOQ, limit of quantification.

Table 3

Intermediate precision and accuracy data.

Compound	Concentration (ng/mL)	Intermediate Precision ^a Analyst A Day 1 (ng/mL)	Analyst B Day 2 (ng/mL)	RSD (%)	Accuracy ^a Average	Standard Deviation	RSE (%)
neochlorogenic acid	Low (3)	3.0	3.0	0.9	404.0	6.0	0.7
	Medium (100)	102.1	102.2	3.9	10,530.0	412.0	2.1
	High (200)	197.7	198.5	2.0	20,367.0	368.0	-0.9

^a Three levels of concentrations ($n = 18$); RSD%, relative standard deviation; RSE%, relative standard error.

Table 4

Compound content of *Ipomea pes-caprae* leaf extract: Expressed as mg/g of extract $\pm$ SD ($n = 3$).

Extract sample	Compound	Content of $\mu\text{g/g}$ extract	RSD (%)
<i>I. pes-caprae</i> leaf extract (20 $\mu\text{g/mL}$)	chlorogenic acid	0.92 ± 0.00	0.31
	neochlorogenic acid	6.07 ± 0.19	3.26
	isochlorogenic acid	0.80 ± 0.04	4.43

material from *Ipomea pes-caprae* leaf.

3.2. In vitro cytotoxicity assay

No significant hemolytic activity was observed to *I. pes-caprae* leaf extract tested, since none of the concentrations used (62.5–1000 $\mu\text{g/mL}$) caused damage to the erythrocyte membrane (Fig. 3), suggesting its low toxicity. In addition, the viability of cells incubated with different treatment concentrations was evaluated through the MTT assay after 24 h of exposure. Extract in the range 31.25–250 $\mu\text{g/mL}$ did not depress the cell viability (NIH/3T3, ATCC® CRL-1658) (Fig. 4), showing

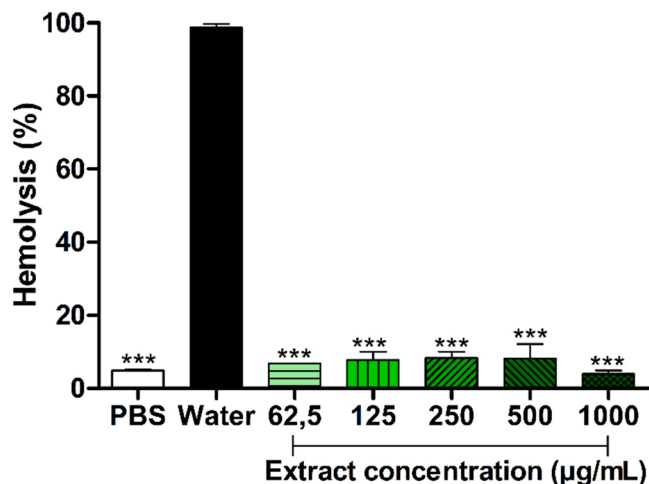


Fig. 3. Hemolytic assay of *I. pes-caprae* leaf extract in different concentrations. Values expressed as mean $\pm$ standard deviation with $n = 3$. *** $p < 0.001$ compared to the positive control (water) and the different concentrations after Tukey's test (ANOVA).

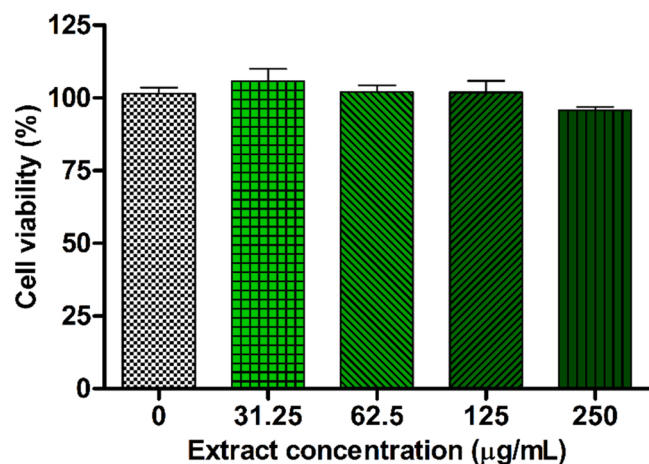


Fig. 4. Cell viability of different concentrations of *Ipomoea pes-caprae* leaf extract against murine fibroblasts (NIH/3T3, ATCC® CRL-1658) in comparison with untreated control after Tukey's test (ANOVA).

absence of cytotoxicity in fibroblast cell, which is of great interest considering the topical use of the extract.

3.3. Physicochemical characterization of formulations

The formulations developed were evaluated macroscopically, after 24 h of preparation, verifying adequate and stable organoleptic characteristics. The developed gel is considered thermosensitive due to its ability to undergo solution-gel transition with increasing temperature, due to the presence of polymer poloxamer F-127 in its composition. The change in the color of the transparent gel, in the case of the placebo, to an orange color in the Ipc gel, was due to the incorporation of the extract, as well as the characteristic odor of each of them. The gels were homogeneous and translucent (Table 5). Fig. 5 shows samples of the base gel (placebo) and the Ipc gel.

The placebo and Ipc gels remained, during the 30 days of observation, without change in color, odor, appearance, or any sign of instability. After the centrifugation test, no changes were seen in the samples, thus showing that the formulations developed were stable.

In the pH assessment (Fig. 6), all formulations showed a pH compatible with the cutaneous pH (4.6–5.8) [41] remaining stable during the 7 days of analysis. These data corroborate with later results obtained in the *in vivo* skin irritation tests.

3.4. Skin irritation test

After evaluating the irritating potential of the gels in topical application in mice, during the 14 days of observation, no changes in the skin, weight, or behavior of the animals were observed, results that corroborate with the pH values presented in the formulations, which were compatible with cutaneous pH.

Table 5

Organoleptic characteristics of the developed gels. Placebo (base gel without extract) and Ipc gel (gel incorporated with extract from leaves of *Ipomoea pes-caprae*).

	Color	Odor	Appearance
Placebo	Transparent	Light	Homogeneous, transparent
Ipc gel	Orange	Characteristic of the extract	Homogeneous, translucent

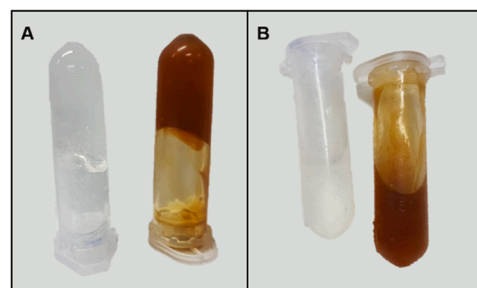


Fig. 5. Gels at 25 °C (A) and 8 °C (B) after 24 h of preparation. In the figures, the placebo (base gel) is shown on the left and the Ipc gel on the right.

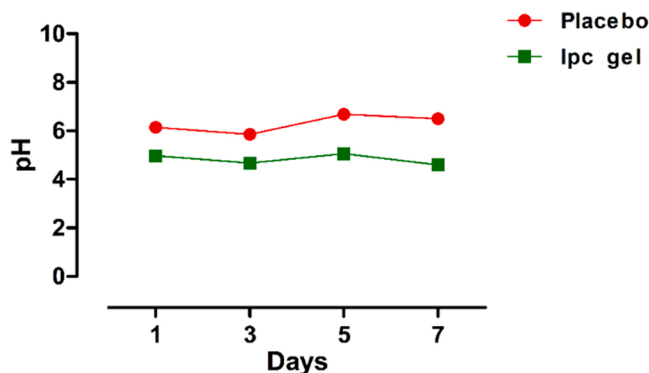


Fig. 6. Evaluation of the pH values of *I. pes-caprae* gels for 7 days after preparation at a temperature of 25 ± 1 °C. Values expressed as mean $\pm$ mean standard error of readings performed in triplicate.

3.5. Acute anti-inflammatory activity

In the model of ear edema induced by croton oil, treatment with the Ipc gel did not show inhibition of the edema and did not lead to a decrease in the levels of the myeloperoxidase enzyme (MPO) as shown in Fig. 7-B.

The anti-inflammatory potential was also tested in the carrageenan-induced paw edema model. The intraplantar injection of carrageenan increased the paw volume of the mice with maximum effect 3 h after the stimulus and treatment with the placebo did not interfere in the installation of the inflammatory process generated. The treatment with the Ipc gel reduced the edema significantly during the observation period (Fig. 8-A), evidently from the first hour after induction. The animals treated with dexamethasone cream (1.0 mg/paw) also had a significant reduction in edema. The area under curve (AUC) analysis clearly showed the antiedematogenic action of the treatments performed (Fig. 8-B). Treatment with Ipc gel and dexamethasone cream significantly reduced AUC (** $p < 0.001$) when compared to control (treatment with placebo, only base gel).

In addition to paw edema, the levels of MPO were assessed. A high activity of MPO was detected after the injection of carrageenan with the placebo treatment, presenting absence of activity in the left paws used as control (without induction of inflammation). The carrageenan-induced MPO activity was significantly inhibited by treatment with dexamethasone (** $p < 0.01$). The use of the Ipc gel in the treatment did not reduce the MPO activity (Fig. 8-C).

3.6. Chronic anti-inflammatory activity

The multi-application model with croton oil allows the installation of an inflammatory process simulating the chronic disease process. Topical treatments with Ipc gel and dexamethasone cream, 9 days after the beginning of the trial and in an already established inflammatory

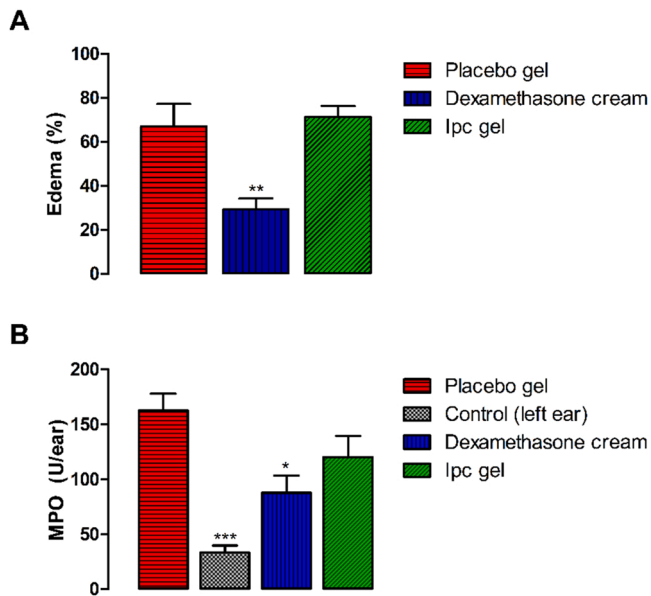


Fig. 7. Topical anti-inflammatory effect of *Ipomoea pes-caprae* leaf extract incorporated in the gel at concentration of 5% on acute edema induced by a single application of croton oil. Ipc gel: gel containing *Ipomoea pes-caprae* leaf extract at 5%. A: Edema was expressed as the percentage difference between the mass of the inflamed right ear and the non-inflamed left ear of each animal. B: Effect of topical application of the gel containing *I. pes-caprae* extract on the levels of myeloperoxidase (MPO). *** $p < 0.001$ and * $p < 0.05$ when compared to the control group (topical treatment with placebo, that is, base gel without extract incorporation) by one-way ANOVA followed by the Tukey test. Values presented as mean $\pm$ standard error of the mean ($n = 5$ /group).

process, presented inhibitions of $47.76 \pm 5.6\%$ and $42.02 \pm 4.8\%$, respectively, significantly reducing ear mass compared to the placebo group (Fig. 9-A). The MPO enzyme activity was reduced in the group treated with Ipc gel (Fig. 9-B).

These results were confirmed by histopathological analysis presented in Fig. 10, where there was a significant increase in the thickness of the dermis accompanied by loosening of the connective tissue due to disorganization of the extracellular matrix fibers, epidermal hyperplasia, and a large number of inflammatory cell infiltrates, when compared to the left ear (not inflamed). The group treated with the Ipc gel was similar to dexamethasone, considerably reducing leukocyte infiltration (Fig. 10-A), edema, and ear thickness (Fig. 10-B), compared to the placebo group.

Based on the results obtained in the evaluation of the levels of tissue thiobarbituric acid reactive substances (expressed in MDA equivalents), it was demonstrated that there was a significant increase in thiobarbituric acid reactive species in the group with chronic inflammatory disease and the treatment performed with Ipc gel or dexamethasone (** $p < 0.01$ when compared to placebo control) produced a reduction in MDA levels (Fig. 10-C).

3.7. Wound healing activity

It was observed that treatment with Ipc gel accelerated the wound closure process, and at the end of the observation time the wound area was also smaller compared to the group that received only the placebo (Fig. 11-A). The difference can be confirmed through the percentage of wound closure, which showed a progressive increase over the days (Fig. 11-B), being on days 3, 5, and 8, respectively, 0.85%, 11.75%, and 57.46% for placebo and 34.99%, 36.11%, and 84.96% for the Ipc gel. These data corroborate the topical anti-inflammatory activity previously observed by Ipc gel in the ear and paw edema models and demonstrate the potential of *I. pes-caprae* in the wound healing process.

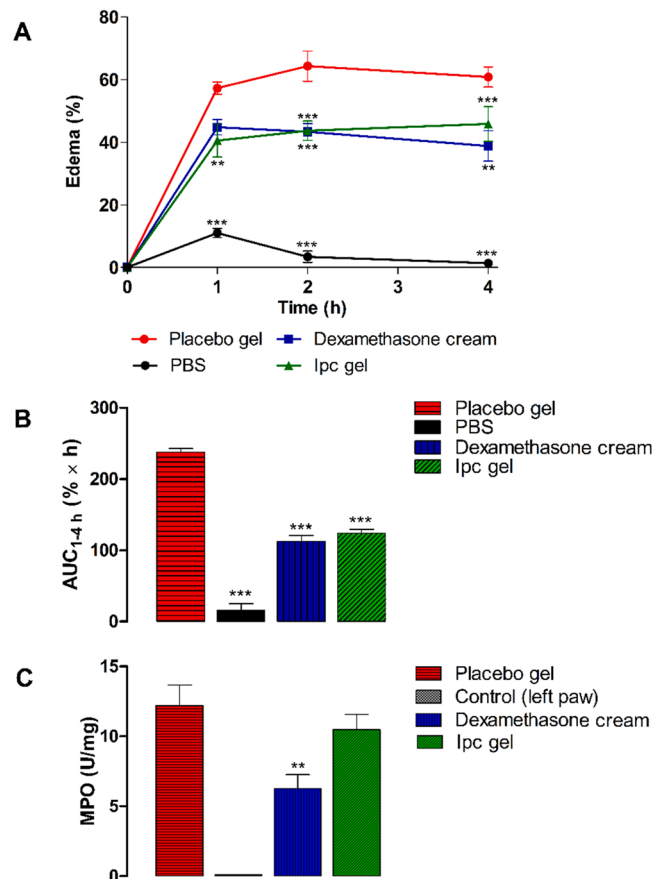


Fig. 8. Topical anti-inflammatory effect of the *Ipomoea pes-caprae* gel on carrageenan-induced paw edema. Ipc gel: gel containing *Ipomoea pes-caprae* leaf extract at 5%. A: Evolution curve of edema during 4 h of experiment. B: Area under the curve (AUC) (0–4 h). C: Effect of topical application of the gel containing *I. pes-caprae* extract on the levels of myeloperoxidase (MPO). Values expressed as mean $\pm$ mean standard error with $n = 5$. ** $p < 0.01$, *** $p < 0.001$ compared to the control group (treatment with placebo, that is, base gel without incorporation of the extract) after Bonferroni test (two-way ANOVA, in A) or Tukey's test (one-way ANOVA, in B). Values presented as mean $\pm$ standard error of the mean ($n = 5$ /group).

Histological analysis showed that on the eighth day, in both placebo and Ipc gel treated animals, there was already tissue without injury, the healing process having started with collagen fibers being deposited, although not yet completely healed. The skin in the group treated with the placebo gel had an intense presence of inflammatory cells (neutrophils), indicating the presence of an acute inflammatory process, and there was no presence of attachments (hairy and sebaceous glands). The skin treated with Ipc gel showed less intensity of inflammatory infiltrate, deposition of young collagen forming, and thin fibers, in addition to the presence of attachments, approaching the appearance of intact skin (Fig. 11-C).

3.8. Inhibition of edema induced by *Bothrops erythromelas* venom

The antiedematogenic evaluation of the Ipc gel, tested in paw edema induced by intraplantar application of *B. erythromelas* venom (BeV), revealed promising data (Fig. 12). The control group treated with the placebo gel showed a significant increase in the animals' paws, as expected. In the time-points observed, treatment with Ipc gel alone was effective in reducing the edema produced by BeV, revealing a significant reduction (** $p < 0.01$) since the first hour analyzed. AV did not show a significant decrease in edema, but its association with Ipc gel proved to be effective in the antiedematogenic response (* $p < 0.05$ and

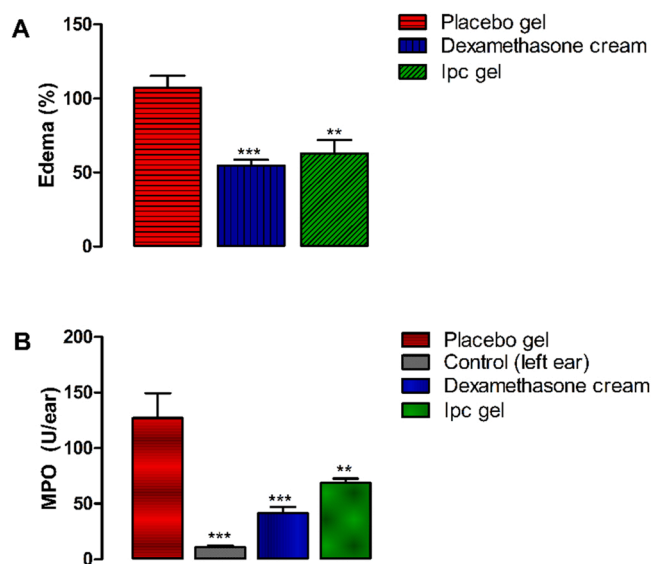


Fig. 9. Topical anti-inflammatory effect of the Ipc gel on chronic edema induced by multiple application of croton oil (% edema). Ipc gel: gel containing *Ipomoea pes-caprae* leaf extract at 5%. A: Edema was expressed as the percentage difference between the mass of the inflamed right ear and the non-inflamed left ear of each animal. B: Effect of topical application of the gel containing *I. pes-caprae* extract on the levels of myeloperoxidase (MPO). *** $p < 0.001$ when compared to the control group (topical treatment with placebo, that is, base gel without extract incorporation) by one-way ANOVA followed by the Tukey test. Values presented as mean $\pm$ standard error of the mean ($n = 5/\text{group}$).

** $p < 0.01$.

4. Discussion

As previously demonstrated in the literature, the hydroethanolic extract of *I. pes-caprae* contains a variety of chemical compounds, such as tannins, saponins, flavonoids, alkaloids, glycosylates, phenolic compounds, unsaturated steroids, and triterpenes [35,39,42]. This diversity

can vary depending on the polarity of the solvent used in the extraction, method used, or agroecological changes [43]. Our results indicate that phenolic compounds are major compounds in the *I. pes-caprae* hydroethanolic leaf extract (Table 1). In this sense, we can highlight *I. pes-caprae* leaf extract as a source of phenolic compounds. This group of metabolites has a remarkable anti-inflammatory capacity due to multiple targets of action such as the inhibition of the production or the action of pro-inflammatory mediators and the gene expression while synthetic drugs generally act in a single way [44–47].

The National Health Surveillance Agency (ANVISA) in Brazil requires that for new herbal medicines it is essential to establish markers to be used in the quality control analysis, from postharvest until final product [48,49]. The marker can be active, when related to the therapeutic activity of the phytocomplex, or analytic, so far as it has no relationship in the therapeutic activity [49]. Knowing the previously described pharmacological potential of phenolic compounds and their large amount in the *I. pes-caprae* leaf extract (observed experimentally in this work and corroborating with previous works in the literature), the quantification was carried out with a view to their application as markers of this raw material. And based on the results, although we cannot conclude that the compounds chlorogenic acid, neochlorogenic acid, and isochlorogenic acid are responsible for the therapeutic action, since the extract has a variety of active compounds that can act synergistically or not contributing to the therapeutic effect the plant could be used as analytical markers for *I. pes-caprae* hydroethanolic leaf extract, being an important step for future studies to evaluate the effect of these compounds or their enriched fractions and to check whether, in addition to being used as analytical markers, they could in fact also be framed as active markers.

Plant extracts are a complex mixture of substances that may or may not act together and present toxic activity, making it important to assess their toxicity to ensure their safety [50]. The *in vitro* cytotoxicity test, evaluated through hemolytic activity, which allows the evaluation of the harmful action of substances on the erythrocyte membrane, and the MTT assay, which allows observing the interference in cell metabolism [24], aim to demonstrate a possible toxicity of the tested substance, being widely used in initial tests in product development. At all doses tested, the *I. pes-caprae* leaf extract did not show any toxicity in the tests carried out, which made it promising for use in pharmacological activity

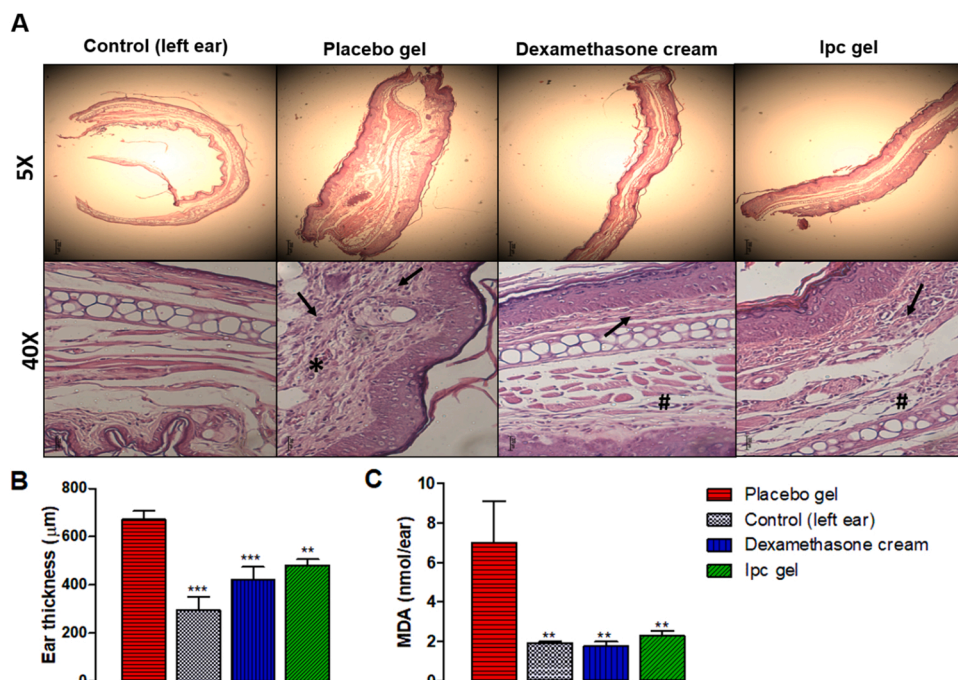


Fig. 10. A: Photomicrograph of mouse ear cross sections after multiple applications of croton oil, stained with hematoxylin-eosin and analyzed under light microscopy (5 and 40x magnification). Arrows indicate the presence of fibroblasts, hashtags indicate the presence of infiltrated inflammatory cells, and asterisks indicate blood vessels. B: The thickness of the edema was expressed as the distance between the inner and outer sides of the ears, measured by analyzing each ear biopsy in 40x magnification using the ImageJ software. C: Effect of topical application of gels containing *Ipomoea pes-caprae* leaf extract at 5.0% on the levels of substances reactive to thiobarbituric acid (SRATs). Ipc: *Ipomoea pes-caprae*. MDA: malondialdehyde. ** $p < 0.01$ and *** $p < 0.001$ when compared to the control group (topical treatment with placebo, that is, base gel without extract incorporation) by one-way ANOVA followed by the Tukey test. Values presented as mean $\pm$ standard error of the mean ($n = 5/\text{group}$).

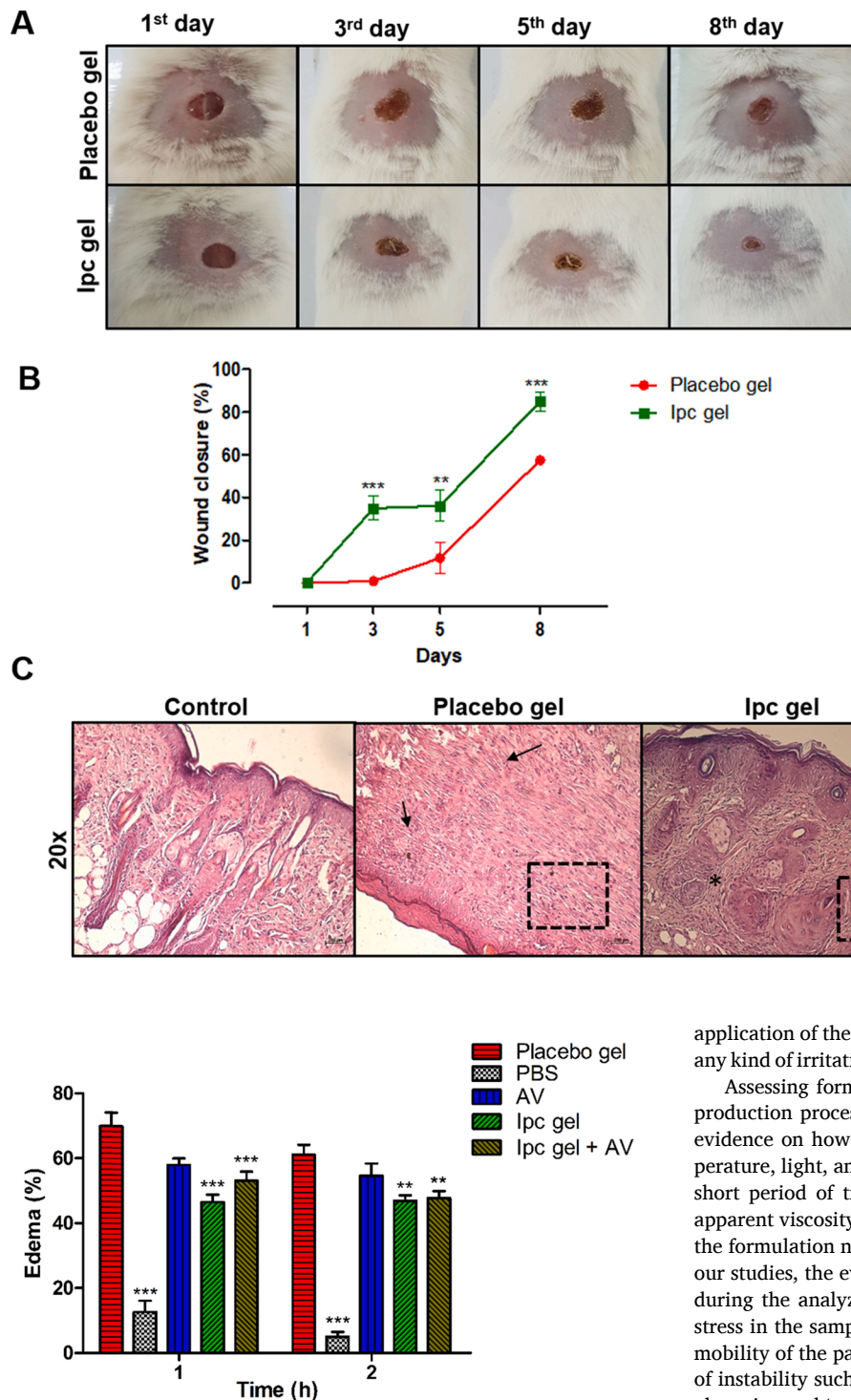


Fig. 12. Inhibition of the edematogenic activity of *B. erythromelas* by Ipc gel. Values were expressed as mean $\pm$ standard error of mean with $n = 5$. ** $p < 0.01$ and *** $p < 0.001$ when compared to the venom control group (Bonferroni's test, two-way ANOVA).

protocols. These results are in agreement with those obtained by Deepak Venkataraman et al. [51], who used the ethanol extract of aerial parts of the plant in an acute dermal toxicity study and obtained an average lethal dose > 2.000 mg/kg, and with Adriano et al. [52], who used the *in vitro* agarose overlay method and observed no skin irritation from the formulations with the extract. These results also corroborate the dermal irritation data of the formulation, where the animals that received the

Fig. 11. A: Monitoring the wound closure process for 7 days in animals treated with Ipc gel. B: Percentage of wound closure. C: Representative photomicrography by cross-sectional optical microscopy of the skin obtained from the healing activity. Hematoxylin-eosin staining at 20x magnification. Arrows represent the presence of inflammatory cells, an asterisk points to skin attachments, and a dotted frame shows deposits of collagen fibers. Values expressed as mean $\pm$ mean standard error with $n = 5$. ** $p < 0.01$, *** $p < 0.001$ compared to the control group (treatment with placebo) after Bonferroni test (two-way ANOVA).

application of the gel containing the *I. pes-caprae* extract did not present any kind of irritation or change in the skin during all the days of analysis.

Assessing formulation stability is of great importance in the initial production process of a drug. These studies are good tools to provide evidence on how the influence of environmental factors such as temperature, light, and humidity can affect the product, obtaining data in a short period of time. Changes in organoleptic characteristics, pH, or apparent viscosity can be detected in these tests, and are indicators that the formulation needs to be evaluated or even reformulated [53,54]. In our studies, the evaluated gel remained stable without physical change during the analyzed time (Table 5). The centrifugation test promotes stress in the sample, simulating an increase in the force of gravity and mobility of the particles, which allows the observation of possible signs of instability such as decomposition, extract precipitation, or any other alteration and/or separation of the components of the formulation [54]. The absence of modifications in this test indicated that the gel could be subjected, without the need for reformulation, to subsequent tests.

During the evaluation of stability parameters, satisfactory results were obtained. No signs of instability were identified such as syneresis, *i. e.*, spontaneous separation of a colloidal system in two phases (gel and liquid), change in color, odor, or the appearance of homogeneity. The pH values remained stable and were similar to that of the skin during the analysis time. Formulations with pH incompatible with skin pH can lead to the development of skin irritation after topical use [41]. These data corroborate with results obtained in the skin irritation test *in vivo*.

The anti-inflammatory test was initially performed in the model of ear edema induced by croton oil. In this model, the inflammatory

process is generated due to the action of phorbol esters present in croton oil, such as 12-O-tetradecanoylphorbol-13-acetate (TPA), which activate protein kinase C (PKC), which in turn participates in the activation of the mitogen-activated protein kinase (MAPK) and phospholipase A₂ pathway. Furthermore, the activation of these pathways leads to the production of interleukins (IL-1, IL-2, IL-6, IL-8), tumor necrosis factor α (TNF- α), and pro-inflammatory enzymes such as cyclooxygenase-2. The participation of these arachidonic acid metabolites is crucial for inflammation to be generated [27,55]. Although previous studies have demonstrated the inhibitory action *in vitro* on prostaglandin synthesis by some compounds that were isolated from the extract of *I. pes-caprae* [56], the treatment with the gel did not show any activity, leading to the hypothesis that the isolated compounds tested may suffer some kind of interference when they are in the complex mixture of the crude extract, or even suffer the action of *in vivo* metabolism, not showing the same activity observed *in vitro*. These results were similar to those previously observed in the literature, where a cream containing Ipc extract was not able to inhibit the edema induced by croton oil, even when applied in different treatment protocols [17,57]. This result may also explain the absence of inhibition in the leukocyte migration, shown indirectly by determining the activity of the myeloperoxidase enzyme, suggesting the lack of extract activity on this inflammatory pathway.

In the carrageenan-induced paw edema model, a test commonly used to screen anti-inflammatory agents for having a two-stage inflammatory process providing assessment over time [58], the Ipc gel significantly decreased the size of the paw edema, similar to the reference drug dexamethasone, with greater intensity in the initial hours. The inflammatory process induced by carrageenan is initially characterized by the release of pro-inflammatory mediators such as kinins, histamine and serotonin, in the first phase (up to 2 h), followed by the release of substances such as prostaglandins, lysozyme, and bradykinin during the second phase [59]. It is expected that phenolic compounds such as derivatives of chlorogenic acid, caffeic acid, and others present in the extract with anti-inflammatory action already tested [17,52,60] participate in the inhibition of the inflammatory pathway induced by kinins (histamine, bradykinin, and serotonin), which may justify the absence of activity in the acute ear edema model, while in the carrageenan-induced edema, especially in the first hour, showed notable inhibition. Some authors have also evidenced the inhibitory action of the Ipc extract on edema induced by histamine and bradykinin, strengthening the hypothesis mentioned above [17]. The action of the gel did not affect the activity of the myeloperoxidase enzyme, demonstrating once again the lack of influence of these compounds on the acute migration of leukocytes.

In addition to the acute inflammation models, the chronic anti-inflammatory activity was also evaluated using the ear edema model induced by multiple applications of croton oil. The multiple applications of croton oil produce, in addition to increased ear mass, the generation of oxidative stress, exudation, and cell infiltration, with the participation mainly of mononuclear leukocytes such as monocytes, macrophages, and T lymphocytes (CD4 + and CD8 +), production of transforming factor β 1 growth factor (TGF- β 1) and production of TNF- α [61]. Thus, it is possible to suggest, at least in part, that the inhibitory action observed by the extract can be the results from inhibition of the activity of cell groups involved in the chronic process, reduction of oxidative stress and the reduction of some interleukins, since in the model of ear edema induced by multiple application of croton oil, the results presented by the gel were promising. The frequency of treatment can also contribute to the results, since in chronic edema the application is carried out twice a day and for 5 consecutive days. Reapplying the gel can contribute to greater exposure to the active ingredients.

Some evidence points out that oxidative stress plays a fundamental role in modulating the release of inflammation mediators, in addition to acting in the expression of adhesion molecules in the endothelium and inflammatory cells, affecting cell recruitment to the inflammation sites [62]. The chronic inflammatory response induced by multiple

applications of croton oil caused an imbalance in the production of reactive oxygen species, resulting in lipid peroxidation and consequently an increase in the concentration of MDA, corroborating previous results [63], thus showing the participation of oxidative stress in the inflammation model. Treatment with the Ipc gel has been revealed to be effective in reducing MDA levels, indicating its potential antioxidant activity. The role of bioactive substances derived from plants in relieving oxidative stress has been widely studied, and phenolic compounds, a major class of the extract of *I. pes-caprae*, have been identified as potent antioxidants [11,45,60,64–67].

The healing of skin wounds is a complex process that involves different steps to restore the integrity of the injured tissue [68]. The inflammatory phase is crucial, as its prolongation may lead to delayed wound closure, which is a worrying factor in clinical practice, as it causes an onerous cost and can contribute to infections and severe consequences for the patient [69]. Therefore, studies have focused on the development of new drugs, mainly based on plants, that promote accelerated healing [70].

Treatment with Ipc gel from the 3rd day of treatment has already shown a significant improvement in wound closure compared to the placebo group, a difference that remained throughout the experiment, and may be associated with the topical anti-inflammatory activity previously shown and demonstrates the potential of *I. pes-caprae* leaf extract in wound healing, contributing to elucidating the popular use of the species. These results agree with those of Muthalib et al. [71], where the effect of the ethanolic extract of *I. pes-caprae* against skin wounds in rabbits were shown, and with those of Eakwaropas et al. [72], who used the Ipc extract incorporated in a hydrogel as a dressing for infected wounds, and both showed efficacy in the treatments. Quinic acid esters (3,5-di-O-caffeoyl-4-O-coumaroylquinic acid and 4,5-di-O-caffeoyl-1,3-di-O-coumaroylquinic acid), compounds identified in *I. pes-caprae* leaf extract, were tested *in vitro* by Teramachi et al. [73] and showed inhibitory activity of collagenase, an enzyme of the matrix metalloproteinase family related to the collagen degradation in the skin, leading to loss of elasticity. In this sense these compounds can be responsible, at least in part, for the wound healing property of *I. pes-caprae* leaf extract. Additionally, the presence of tannins and flavonoids, such as quercetin, isoquercetin, and isoquercitrin, in the chemical composition of the extract can aid in wound healing by presenting antioxidant activity, promoting wound contraction, and increasing capillary formation and fibroblast proliferation [74,75], directly contributing to the results previously presented.

I. pes-caprae is often used in folk medicine to treat snake bites [14,76] and this study allowed us to evaluate the effect of the gel on the edema induced by the snake venom *B. erythromelas*. The edema developed in the envenomation by *Bothrops* snakes is related to several factors that lead to the release of toxins such as phospholipases A₂ and metalloproteases present in the venom, inducing changes in the microvasculature and increased permeability of capillaries and venules, in addition to the indirect participation of pro-inflammatory mediators released at the site, such as histamine, prostaglandins, kinins, and the complement system [77,78].

Antiedematogenic evaluation exhibited the efficacy of treatment using Ipc gel and its association with AV in reducing edema induced by *B. erythromelas*, whereas treatment with AV alone was not effective. The Ipc gel was also able to inhibit the edema induced by *B. jararaca* (data not shown). It is noteworthy that serum therapy, despite being the only standardized treatment for snakebite accidents, has a limitation in inhibiting the local effects caused by the venom, mainly in cases where the bite-treatment time is prolonged, which is a remarkable fact since most of those affected live far from health centers. The low efficacy of AV is related to the fact that it acts only by neutralizing the toxins present in the venom, not preventing the effects triggered after their initial action nor the action of endogenous inflammatory mediators [79].

Thus, using Ipc extract to complement AV serum therapy proved to

be effective in reducing edema, with the possibility of acting directly on toxins, as it has phenolic compounds in its composition, for example, which have, in addition to antioxidant action, metal chelators (essential for catalytic activity of toxins) [80]. This inhibition can also be attributed to the anti-inflammatory properties of the extract on the released endogenous inflammatory mediators.

The literature also points to the marked participation of mast cells in bothropic edema induced by *Bothrops moojeni* and, as the inhibitory action of treatment with *I. pes-caprae* in histamine-induced edema has been presented in the literature [17], there is even more evidence of how promising this therapy is. It is noteworthy that another very relevant local effect in bothropic envenomation, dermonecrosis, could also benefit the treatment with the gel containing the extract of *I. pes-caprae*, taking into account that the potent healing activity of the extract was observed.

Given the above, further studies are needed to better understand the mechanisms involved with these effects, since the gel containing the extract of *I. pes-caprae* proves to be an interesting and promising pharmaceutical formulation, with potential for the treatment of topical inflammatory skin diseases and as a complementary therapy for edema induced by bothropic venom.

5. Conclusions

The results presented in this approach provide information about the potential of a gel containing *I. pes-caprae* leaf extract as a promising, safe, and effective formulation to be used in the treatment of inflammatory skin diseases and wound healing. In addition, the antiedematogenic action in envenoming by bothropic snakes, demonstrated for the first time in the literature, supports the popular use of the plant species. These effects were mainly observed through increasing the antioxidant defense system and modulating inflammatory markers. Overall, this work supports the further development of a new pharmaceutical product or a new source of raw material as an active ingredient of a topical anti-inflammatory gel.

Funding

This research was supported by grants from CAPES (Grant number: 88882376258/2019-01), JBXS thanks CAPES for the Ph.D. scholarship. M.F. Fernandes-Pedrosa and A.A. Silva-Junior are CNPq fellowship-honored researchers. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Credit authorship contribution statement

JBXS, JFS, AASJ, and MFFP designed the experiments. JBXS, JFS, JGRP, JASG, and JVCC conducted the collection of plant material, preparation of plant extracts, and *in vivo* experiments. JBXS, JSFA, RMS, and NPL conducted the phytochemical analysis of plant extract. JBXS, JGRP, and AASJ developed and conducted the physicochemical characterization of formulations. VBG and JBXS performed the histopathological analysis. JBXS, JFS, VBG, RFAJ, JSFA, RMS, NPL, SMZ, AASJ, and MFFP analyzed the data. RFAJ, NPL, SMZ, AASJ, and MFFP contributed with reagents, materials, and/or analysis. JBXS wrote the manuscript. RFAJ, SMZ, AASJ, JFS, and MFFP refined the manuscript for publication. All authors have read and agreed to the published version of the manuscript.

Conflict of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank all participants for their valuable time and commitment to the study. We also thank the Federal University of Rio Grande do Norte for its support in the development of this research.

References

- [1] M.J. Ahmed, G. Murtaza, A study of medicinal plants used as ethnoveterinary: harnessing potential phytotherapy in Bheri, District Muzaffarabad (Pakistan), *J. Ethnopharmacol.* 159 (2015) 209–214, <https://doi.org/10.1016/j.jep.2014.11.016>.
- [2] M.A.M. Maciel, A.C. Pinto, V.F.V. Junior, N.F. Grynberg, A. Echevarria, Plantas medicinais: a necessidade de estudos multidisciplinares, *Quim. Nova* 25 (2002) 429–438, <https://doi.org/10.1590/S0100-40422002000300016>.
- [3] S. Hayta, S. Selvi, Traditional uses of medicinal plants in Elazığ (Turkey), *J. Ethnopharmacol.* 154 (2014) 613–623, <https://doi.org/10.1016/j.jep.2014.04.026>.
- [4] WHO, WHO Guidelines for Assessing Quality of Herbal Medicines with Reference to Contaminants and Residues, 2007, p. 105. (<https://apps.who.int/iris/handle/10665/43510>).
- [5] R.C. Dutra, M.M. Campos, A.R.S. Santos, J.B. Calixto, Medicinal plants in Brazil: pharmacological studies, drug discovery, challenges and perspectives, *Pharmacol. Res.* 112 (2016) 4–29, <https://doi.org/10.1016/j.phrs.2016.01.021>.
- [6] P.M. Leite, L.M. Camargos, R.O. Castilho, Recent progress in phytotherapy: a Brazilian perspective, *Eur. J. Integr. Med.* 41 (2021), 101270, <https://doi.org/10.1016/j.eujim.2020.101270>.
- [7] K. Barnes, L. Ball, B. Desbrow, N. Alsharairi, F. Ahmed, Consumption and reasons for use of dietary supplements in an Australian university population, *Nutrition* 32 (2016) 524–530, <https://doi.org/10.1016/j.nut.2015.10.022>.
- [8] A.N. Welz, A. Emberger-klein, K. Menrad, Why people use herbal medicine: insights from a focus-group study in Germany, *BMC Complement. Altern. Med.* (2018) 1–9, <https://doi.org/10.1186/s12906-018-2160-6>.
- [9] E.W.C. Chan, S. Baba, H.T. Chan, M. Kainuma, J. Tangah, Medicinal plants of sandy shores: a short review on *Vitex trifolia* L. and *Ipomoea pes-caprae* (L.) R. Br., *Indian, J. Nat. Prod. Resour.* 7 (2016) 107–115.
- [10] M. Meira, E. Pereira, J.M. David, J.P. David, Review of the genus *Ipomoea*: traditional uses, chemistry and biological activities, *Braz. J. Pharmacogn.* 22 (2012) 682–713, <https://doi.org/10.1590/S0102-695X2012005000025>.
- [11] P. Giovannini, M.J.R. Howes, Medicinal plants used to treat snakebite in Central America: review and assessment of scientific evidence, *J. Ethnopharmacol.* 199 (2017) 240–256, <https://doi.org/10.1016/j.jep.2017.02.011>.
- [12] O. Neamsuvan, P. Bunmee, A survey of herbal weeds for treating skin disorders from Southern Thailand: Songkhla and Krabi Province, *J. Ethnopharmacol.* 193 (2016) 574–585, <https://doi.org/10.1016/j.jep.2016.09.048>.
- [13] D. Vieira, C. Padoani, J. dos, S. Soares, J. Adriano, V.C. Filho, M.M. de Souza, T.M. B. Bresolin, A.G. Couto, Development of hydroethanolic extract of *Ipomoea pes-caprae* using factorial design followed by antinociceptive and anti-inflammatory evaluation, *Braz. J. Pharmacogn.* 23 (2013) 72–78, <https://doi.org/10.1590/S0102-695X2012005000126>.
- [14] F.G. Coe, G.J. Anderson, Snakebite ethnopharmacopoeia of eastern Nicaragua, *J. Ethnopharmacol.* 96 (2005) 303–323, <https://doi.org/10.1016/j.jep.2004.09.026>.
- [15] A. Manigaunha, N. Ganesh, M.D. Kharya, Morning glory: a new thirst in-search of de-novo therapeutic approach, *Int. J. Phytomed.* 2 (2010) 18–21, <https://doi.org/10.5138/ijpm.2010.0975.0185.02003>.
- [16] R. Nilam, P. Jyoti, C. Sumitra, Pharmacognostic and phytochemical studies of *Ipomoea pes-caprae*, an halophyte from Gujarat, *J. Pharmacogn. Phytochem.* 7 (2018) 11–18.
- [17] C.S. Barth, H.G. Tolentino de Souza, L.W. Rocha, G.F. da Silva, M.F. dos Anjos, V. D. Pastor, T.M. Belle Bresolin, A. Garcia Couto, J. Roberto Santin, N.L. Meira Quintão, *Ipomoea pes-caprae* (L.) R. Br (Convolvulaceae) relieved nociception and inflammation in mice – a topical herbal medicine against effects due to cnidarian venom-skin contact, *J. Ethnopharmacol.* 200 (2017) 156–164, <https://doi.org/10.1016/j.jep.2017.02.014>.
- [18] N.D. Venkataraman, W.C. Atleeb, T.P. Prabhuc, R. Kannana, Anti-inflammatory potential of ethanolic extracts from aerial parts of *Ipomoea pes-caprae* (L.) R.Br using cotton pellet induced granuloma model, *J. Appl. Pharm. Sci.* 3 (2013) 61–63, <https://doi.org/10.7324/JAPS.2013.3711>.
- [19] J.S.F. Alves, J.I. Marques, D.P. Demarque, L.R.F. Costa, J.G. Amaral, N.P. Lopes, A. A. da Silva-Júnior, L.A.L. Soares, E.C. Gavioli, L.D.S. Ferreira, S.M. Zucolotto, Involvement of isoorientin in the antidepressant bioactivity of a flavonoid-rich extract from *Passiflora edulis* f. *flavicarpa* leaves, *Rev. Bras. Farmacogn.* 30 (2020) 240–250, <https://doi.org/10.1007/s43450-020-00003-x>.
- [20] Z.Y. Gu, C.Y. Feng, S.S. Li, D.D. Yin, Q. Wu, L. Zhang, L.S. Wang, Identification of flavonoids and chlorogenic acids in elm fruits from the genus *Ulmus* and their antioxidant activity, *J. Sep. Sci.* 42 (2019) 2888–2899, <https://doi.org/10.1002/jssc.201900302>.
- [21] ICH, Validation of analytical procedures: text and methodology Q2(R1), in: Proceedings of the International Conference on Harmonisation (ICH) ICH Topic Q2 (R1) Validation of Analytical Procedures: Text and Methodology, 2005, pp. 1–13.
- [22] E.A. Moreira, A.C. Pilon, L.E. Andrade, N.P. Lopes, New perspectives on chlorogenic acid accumulation in harvested leaf tissue: impact on traditional

- medicine preparations, ACS Omega 3 (2018) 18380–18386, <https://doi.org/10.1021/acsomega.8b02409>.
- [23] C. Gauthier, J. Legault, C. Girard-lancette, V. Mshvildadze, A. Pichette, Haemolytic activity, cytotoxicity and membrane cell permeabilization of semi-synthetic and natural lupane- and oleanane-type saponins, Bioorg. Med. Chem. 17 (2009) 2002–2008, <https://doi.org/10.1016/j.bmc.2009.01.022>.
- [24] T. Mosmann, Rapid colorimetric assay for cellular growth and survival: application to proliferation and cytotoxicity assays, J. Immunol. Methods 65 (1983) 55–63, [https://doi.org/10.1016/0022-1759\(83\)90303-4](https://doi.org/10.1016/0022-1759(83)90303-4).
- [25] J.B. Xavier-Santos, J. Félix-Silva, J.G.R. Passos, J.A.S. Gomes, J.M. Fernandes, V. B. Garcia, R.F. de Araujo-Junior, S.M. Zucolotto, A.A. Silva-Junior, M.F. Fernandes-Pedrosa, Development of an effective and safe topical anti-inflammatory gel containing *Jatropha gossypifolia* leaf extract: results from a pre-clinical trial in mice, J. Ethnopharmacol. 227 (2018), <https://doi.org/10.1016/j.jep.2018.09.007>.
- [26] I. Posadas, M. Bucci, F. Roviezzo, A. Rossi, L. Parente, L. Sautebin, G. Cirino, Carrageenan-induced mouse paw oedema is biphasic, age-weight dependent and displays differential nitric oxide cyclooxygenase-2 expression, Br. J. Pharmacol. (2004) 331–338, <https://doi.org/10.1038/sj.bjp.0705650>.
- [27] P.L. Stanley, S. Steiner, M. Havens, K.M. Trampusch, Mouse skin inflammation induced by multiple topical applications of 12-O-tetradecanoylphorbol-13-acetate, Skin Pharm. 4 (1991) 262–271, <https://doi.org/10.1159/000210960>.
- [28] J.A. Buege, S.D. Aust, Microsomal lipid peroxidation, Methods Enzymol. 52 (1978) 302–310, [https://doi.org/10.1016/S0076-6879\(78\)52032-6](https://doi.org/10.1016/S0076-6879(78)52032-6).
- [29] M.M. Bradford, A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding, Anal. Biochem. 72 (1976) 248–254, [https://doi.org/10.1016/0003-2697\(76\)90527-3](https://doi.org/10.1016/0003-2697(76)90527-3).
- [30] J. Félix-Silva, J.A.S. Gomes, J.B. Xavier-Santos, J.G.R. Passos, A.A. Silva-Junior, D. V. Tambourgi, M.F. Fernandes-Pedrosa, Inhibition of local effects induced by *Bothrops erythromelas* snake venom: assessment of the effectiveness of Brazilian polyvalent bothropic antivenom and aqueous leaf extract of *Jatropha gossypifolia*, Toxicon 125 (2017), <https://doi.org/10.1016/j.toxicon.2016.11.260>.
- [31] F. Manhães, B. Gonçalves, A.C. Ramos, S. Mathias, Q.D.S. Sales, C.C. Ramos, F. Antunes, R.R. De Oliveira, Phytochemical analysis and hypotensive activity of *Ipomoea pes-caprae* on blood pressure of normotensive rats, Pharmacognosy (2020), <https://doi.org/10.1590/2175-7860202071048>.
- [32] R. Jaiswal, M.A. Patras, P.J. Eravuchira, N. Kuhnert, Profile and characterization of the chlorogenic acids in green *Robusta coffea* beans by LC-MSn: identification of seven new classes of compounds, J. Agric. Food Chem. 58 (2010) 8722–8737, <https://doi.org/10.1021/jf1014457>.
- [33] U. Lawal, S.W. Leong, K. Shaari, I.S. Ismail, A. Khatib, F. Abas, α -Glucosidase inhibitory and antioxidant activities of different *Ipomoea aquatica* cultivars and LC-MS/MS profiling of the active cultivar, J. Food Biochem. 41 (2017), <https://doi.org/10.1111/jfbc.12303>.
- [34] M.N. Clifford, S. Knight, B. Surucu, N. Kuhnert, Characterization by LC-MS n of four new classes of chlorogenic acids in green coffee beans: dimethoxycinnamoylquinic acids, diferuloylquinic acids, caffeoyl-dimethoxycinnamoylquinic acids, and feruloyl-dimethoxycinnamoylquinic acids, J. Agric. Food Chem. 54 (2006) 1957–1969, <https://doi.org/10.1021/jf0601665>.
- [35] D.M. Dutra, C. Da Silva Barth, L.C. Block, N.L.M. Quintão, A.G. Couto, V. Cechinel Filho, T.M.B. Bresolin, Simultaneous determination of four phenolic compounds in extracts of aerial parts of *Ipomoea pes-caprae* (L.) R. Br. (Convolvulaceae) by HPLC-UV, Quim. Nova 37 (2014) 1510–1514, <https://doi.org/10.5935/0100-4042.20140237>.
- [36] R. Ben Said, A.I. Hamed, U.A. Mahalel, A.S. Al-Ayed, M. Kowalczyk, J. Moldoch, W. Oleszek, A. Stochmal, Tentative characterization of polyphenolic compounds in the male flowers of *Phoenix dactylifera* by liquid chromatography coupled with mass spectrometry and DFT, Int. J. Mol. Sci. 18 (2017) 1–18, <https://doi.org/10.3390/ijms18030512>.
- [37] A.A. Chernonosov, E.A. Karpova, E.M. Lyakh, Identification of phenolic compounds in *Myricaria bracteata* leaves by high-performance liquid chromatography with a diode array detector and liquid chromatography with tandem mass spectrometry, Braz. J. Pharmacogn. 27 (2017) 576–579, <https://doi.org/10.1016/j.bjp.2017.07.001>.
- [38] A. Wang, R. Li, L. Ren, X. Gao, Y. Zhang, Z. Ma, D. Ma, Y. Luo, A comparative metabolomics study of flavonoids in sweet potato with different flesh colors (*Ipomoea batatas* (L.) Lam), Food Chem. 260 (2018) 124–134, <https://doi.org/10.1016/j.foodchem.2018.03.125>.
- [39] C. da, S. Barth, H.G.T. de Souza, L.W. Rocha, C.R. de, S. Madeira, C. Assis, T. Bonomini, A. Malheiros, L.P. Sandjo, R. Niero, C. Filho, A.G. Couto, J.R. Santin, N.L.M. Quintão, T.M.B. Bresolin, RP-HPLC and LC-MS-MS determination of a bioactive artefact from *Ipomoea pes-caprae* extract, Braz. J. Pharmacogn. 29 (2019) 570–577, <https://doi.org/10.1016/j.bjp.2019.05.005>.
- [40] D.P. Demarque, A.E.M. Crotti, R. Vesecchi, J.L.C. Lopes, N.P. Lopes, Fragmentation reactions using electrospray ionization mass spectrometry: an important tool for the structural elucidation and characterization of synthetic and natural products, Nat. Prod. Rep. 33 (2016) 432–455, <https://doi.org/10.1039/C5NP00073D>.
- [41] G.R. Leonardi, L.R. Gaspar, P.M.B.G. Maia Campos, Estudo da variação do pH da pele humana exposta à formulação cosmética acrescida ou não das vitaminas A, E ou de ceramida, por metodologia não invasiva, An. Bras. Dermatol. 77 (2002) 563–569, <https://doi.org/10.1590/S0365-05962002000500006>.
- [42] P. Nagababu, U. Rao, Pharmacological potential of *Ipomoea pes-caprae* (L.) R. Br. whole plant extracts, Der Pharm. Sin. 6 (2015) 52–60, <https://doi.org/10.1590/S0102-695X2009000600012>.
- [43] C. da Silva Barth, H.G. Tolentino de Souza, L.W. Rocha, G.F. da Silva, M.F. dos Anjos, V.D. Pastor, T.M. Belle Bresolin, A. Garcia Couto, J. Roberto Santin, N.L. Meira Quintão, *Ipomoea pes-caprae* (L.) R. Br. (Convolvulaceae) relieved nociception and inflammation in mice – a topical herbal medicine against effects due to cutaneous venom-skin contact, J. Ethnopharmacol. 200 (2017) 156–164, <https://doi.org/10.1016/j.jep.2017.02.014>.
- [44] A.A. Furtado, M. Torres-Rêgo, M.C.J.S. Lima, M.A.O. Bitencourt, A.B. Estrela, N. Souza da Silva, E.M. da Silva Siqueira, J.C. Tomaz, N.P. Lopes, A.A. Silva-Junior, S.M. Zucolotto, M.F. Fernandes-Pedrosa, Aqueous extract from *Ipomoea asarifolia* (Convolvulaceae) leaves and its phenolic compounds have anti-inflammatory activity in murine models of edema, peritonitis and air-pouch inflammation, J. Ethnopharmacol. 192 (2016) 225–235, <https://doi.org/10.1016/j.jep.2016.07.048>.
- [45] V. Alagesan, S. Ramalingam, M. Kim, S. Venugopal, Antioxidant activity guided isolation of a coumarin compound from *Ipomoea pes-caprae* (Convolvulaceae) leaves acetone extract and its biological and molecular docking studies, Eur. J. Integr. Med. 32 (2019), 100984, <https://doi.org/10.1016/j.eujim.2019.100984>.
- [46] R. Hussein, A.A. El-Anssary, Plants secondary metabolites: the key drivers of the pharmacological actions of medicinal plants, Herb. Med. (2018) (books.google.com).
- [47] D.L. Ambriz-Pérez, N. Leyva-López, E.P. Gutierrez-Grijalva, J.B. Heredia, Phenolic compounds: natural alternative in inflammation treatment. A review, Cogent Food Agric. 2 (2016), 1131412, <https://doi.org/10.1080/23311932.2015.1131412>.
- [48] A. Cecília, B. Carvalhal, L.S. Ramalho, R. France, D.O. Marques, Regulation of herbal medicines in Brazil, J. Ethnopharmacol. 2013 (2014) 1–4, <https://doi.org/10.1016/j.jep.2014.08.019>.
- [49] BRASIL, Ministério da Saúde Resolução da diretoria colegiada - RDC No. 26, DE 13 DE MAIO DE, in: Ministério Da Saúde. Resolução da dir. colegiada - RDC No. 26, 13 MAIO 2014. Dispõe Sobre o Regist. Medicam. Fitoterápicos e o Regist. e a Notif. Prod. Tradic. Fitoterápicos, 2014, p. 34.
- [50] T. Alelign, D. Chalchisa, N. Fekadu, D. Solomon, T. Sisay, A. Debella, B. Petros, Evaluation of acute and sub-acute toxicity of selected traditional antiulcerogenic medicinal plant extracts in Wistar albino rats, Toxicol. Rep. 7 (2020) 1356–1365, <https://doi.org/10.1016/j.toxrep.2020.10.001>.
- [51] N. Deepak Venkataraman, W. Clement Atlee, P. Muralidharan, T. Purushoth Prabhu, M.S. Priya, S. Muthukumar, Assessment of acute dermal toxicity of ethanolic extracts from aerial parts of *Ipomoea pes-caprae* (L.) R. Br on wistar albino rats, Res. J. Pharm. Biol. Chem. Sci. 4 (2013) 769–776.
- [52] J. Adriano, D.J.A. Netz, T.M.B. Bresolin, R.A. de Freitas, A.G. Couto, Development of a topical gel containing dried extract of *Ipomoea pes-caprae brasiliensis* (L.) R. Br. (Convolvulaceae), J. Med. Plants Res. 8 (2014) 1074–1080, <https://doi.org/10.5897/JMPR2014.5466>.
- [53] R.M. Daudt, P.I. Back, N.S.M. Cardozo, L.D.F. Marczak, I.C. Külkamp-Guerreiro, Pinhão starch and coat extract as new natural cosmetic ingredients: Topical formulation stability and sensory analysis, Carbohydr. Polym. 134 (2015) 573–580, <https://doi.org/10.1016/j.carbpol.2015.08.038>.
- [54] BRASIL, Ministério da Saúde, Guia de Estabilidade de Produtos Cosméticos, Guia Estabilidade Prod. Cosméticos, 2004, p. 49.
- [55] M. Conceição de Souza, J.T. Hubner, G.M. Gonçalves, T. Ferreira da Costa Fernandes, R. Laureano-Melo, W. da Silva Côrtes, L. Gonçalves, F. de Assis da Silva, B.G. Marinho, Topical and systemic use of *Joannesia princeps* Vell. LC seed oil in acute pain and inflammation induced by different agents, J. Ethnopharmacol. 268 (2021), 113554, <https://doi.org/10.1016/j.jep.2020.113554>.
- [56] U. Pongprayoon, P. Baekström, U. Jacobsson, M. Lindström, L. Bohlin, Compounds inhibiting prostaglandin synthesis isolated from *Ipomoea pes-caprae*, Planta Med. 57 (1991) 515–518, <https://doi.org/10.1055/s-2006-960196>.
- [57] C.S. Barth, Desenvolvimento de formulação semissólida contendo extrato de *Ipomoea pes-caprae* (L.) R. Br. (convolvulaceae) e avaliação da atividade anti-inflamatória e antinociceptiva, Universidade do Vale do Itajaí, 2014.
- [58] A. Rauf, G. Uddin, B.S. Siddiqui, A. Khan, H. Khan, M. Arfan, N. Muhammad, A. Wadood, In-vivo antinociceptive, anti-inflammatory and antipyretic activity of pistagreed acid isolated from *Pistacia integerrima*, Phytomedicine 21 (2014) 1509–1515, <https://doi.org/10.1016/j.phymed.2014.07.015>.
- [59] R.R. Brooks, J.F. Carpenter, S.M. Jones, T.C. Ziegler, S.F. Pong, Canine carrageenin-induced acute paw inflammation model and its response to nonsteroidal antiinflammatory drugs, J. Pharmacol. Methods 25 (1991) 275–283, [https://doi.org/10.1016/0160-5402\(91\)90027-3](https://doi.org/10.1016/0160-5402(91)90027-3).
- [60] D. Banerjee, A.K. Hazra, T. Seal, T. Sur, D. Bhattacharya, J. Ray, A. Mukherjee, B. Mukherjee, Antioxidant and antiinflammatory activities of different solvent extracts and isolated compounds of *Ipomoea pes-caprae* (L.) sweet of sunderban mangrove eco-complex, Asian J. Chem. 25 (2013) 4997–5000, <https://doi.org/10.14233/ajchem.2013.14226>.
- [61] W.M. Cruvinel, D.M. Júnior, J.A.P. Araújo, T.T.T. Catelan, A.W.S.S. Souza, N. P. Silva, L.E.C. Andrade, Sistema Imunitário – parte I Fundamentos da imunidade inata com ênfase nos mecanismos moleculares e celulares da resposta inflamatória, Rev. Bras. Reumatol. 50 (2010) 434–461, <https://doi.org/10.1590/S0482-50042010000400008>.
- [62] B. Halliwell, M. Whiteman, Measuring reactive species and oxidative damage in vivo and in cell culture: how should you do it and what do the results mean? Br. J. Pharmacol. 142 (2004) 231–255, <https://doi.org/10.1038/sj.bjp.0705776>.
- [63] A. Djordjević, P. Kotnik, D. Horvat, Ž. Knez, M. Antonić, Pharmacodynamics of malondialdehyde as indirect oxidative stress marker after arrested-heart cardiopulmonary bypass surgery, Biomed. Pharmacother. 132 (2020) 1–5, <https://doi.org/10.1016/j.biopha.2020.110877>.
- [64] V. Alagesan, S. Ramalingam, M. Kim, S. Venugopal, Antioxidant activity guided isolation of a coumarin compound from *Ipomoea pes-caprae* (Convolvulaceae) leaves acetone extract and its biological and molecular docking studies, Eur. J. Integr. Med. 32 (2019), 100984, <https://doi.org/10.1016/j.eujim.2019.100984>.

- [65] M. Qasim, Z. Abideen, M.Y. Adnan, S. Gulzar, B. Gul, M. Rasheed, M.A. Khan, Antioxidant properties, phenolic composition, bioactive compounds and nutritive value of medicinal halophytes commonly used as herbal teas, *S. Afr. J. Bot.* 110 (2017) 240–250, <https://doi.org/10.1016/j.sajb.2016.10.005>.
- [66] A. Kumar, S. Paul, P. Kumari, S. Thirugnanasambandan Somasundaram, K. Kathiresan, Antioxidant and free radical scavenging activities of *Ipomoea pes-caprae* (L.) R. Br. Extracts, *Int. J. Curr. Pharm. Rev. Res.* 5 (2015) 91–109.
- [67] C.M. Rodríguez-García, J.C. Ruiz-Ruiz, L. Peraza-Echeverría, S.R. Peraza-Sánchez, L.W. Torres-Tapia, D. Pérez-Brito, R. Tapia-Tussell, F.G. Herrera-Chalé, M. R. Segura-Campos, A. Quijano-Ramayo, J.M. Ramón-Sierra, E. Ortiz-Vázquez, Antioxidant, antihypertensive, anti-hyperglycemic, and antimicrobial activity of aqueous extracts from twelve native plants of the Yucatan coast, *PLoS One* 14 (2019) 1–17, <https://doi.org/10.1371/journal.pone.0213493>.
- [68] A. Bhaskar Rao, E. Prasad, S. Siva Deepthi, V. Haritha, S. Ramakrishna, K. Madhusudan, M. Venkata Surekha, Y. Sri Rama Venkata Rao, Wound healing: a new perspective on glucosylated tetrahydrocurcumin, *Drug Des. Dev. Ther.* 9 (2015) 3579–3588, <https://doi.org/10.2147/DDDT.S85041>.
- [69] I. Negut, V. Grumezescu, A.M. Grumezescu, Treatment strategies for infected wounds, *Molecules* 23 (2018) 1–23, <https://doi.org/10.3390/molecules23092392>.
- [70] A. Shedeova, D. Leavesley, Z. Upton, C. Fan, Wound healing and the use of medicinal plants, *Evid. Based Complement. Altern. Med.* 2019 (2019), <https://doi.org/10.1155/2019/2684108>.
- [71] E.M. Muthalib, Fatimawali, H.J. Edy, Formulasi Salep Ekstrak Etanol Daun Tapak Kuda (*Ipomoea pes-caprae*) Dan Uji Efektivitasnya Terhadap Luka Terbuka Pada Punggung Kelinci, *Pharmakon* 2 (2013) 79–82, <https://doi.org/10.35799/pha.2.2013.2384>.
- [72] P. Eakwaropas, T. Ngawhirunpat, T. Rojanarata, P. Akkaramongkolporn, P. Opanasopit, P. Patrojanasophon, Fabrication of electrospun hydrogels loaded with *Ipomoea pes-caprae* (L.) R. Br extract for infected wound, *J. Drug Deliv. Sci. Technol.* 55 (2020), 101478, <https://doi.org/10.1016/j.jddst.2019.101478>.
- [73] F. Teramachi, T. Koyano, T. Kowithayakorn, M. Hayashi, K. Komiya, M. Ishibashi, Collagenase inhibitory quinic acid esters from *Ipomoea pes-caprae*, *J. Nat. Prod.* 68 (2005) 794–796, <https://doi.org/10.1021/np0500631>.
- [74] X. Su, X. Liu, S. Wang, B. Li, T. Pan, D. Liu, F. Wang, Y. Diao, K. Li, Wound-healing promoting effect of total tannins from *Entada phaseoloides* (L.) Merr. in rats, *Burns* 43 (2017) 830–838, <https://doi.org/10.1016/j.burns.2016.10.010>.
- [75] E. Yadav, D. Singh, P. Yadav, A. Verma, Antioxidant and anti-inflammatory properties of *Prosopis cineraria* based phenolic rich ointment in wound healing, *Biomed. Pharmacother.* 108 (2018) 1572–1583, <https://doi.org/10.1016/j.biopha.2018.09.180>.
- [76] P. Giovannini, M.R. Howes, Medicinal plants used to treat snakebite in Central America: review and assessment of scientific evidence, *J. Ethnopharmacol.* 199 (2017) 240–256, <https://doi.org/10.1016/j.jep.2017.02.011>.
- [77] J. Gutiérrez, B. Lomonte, Phospholipase A2 myotoxins from *Bothrops* snake venoms, *Toxicon* 33 (1995) 1405–1424, [https://doi.org/10.1016/0041-0101\(95\)00085-Z](https://doi.org/10.1016/0041-0101(95)00085-Z).
- [78] J.M. Gutiérrez, A. Rucavado, F. Chaves, C. Díaz, T. Escalante, Experimental pathology of local tissue damage induced by *Bothrops asper* snake venom, *Toxicon* 54 (2009) 958–975, <https://doi.org/10.1016/j.toxicon.2009.01.038>.
- [79] J.M. Gutierrez, G. Leoá, G. Rojas, B. Lomonte, A. Rucavado, F. Chaves, Neutralization of local tissue damage induced by *Bothrops asper* (Terciopelo) Snake Venom, *Toxicon* 36 (1998) 1529–1538, [https://doi.org/10.1016/S0041-0101\(98\)00145-7](https://doi.org/10.1016/S0041-0101(98)00145-7).
- [80] W.B. Mors, M.C. do Nascimento, B.M.R. Pereira, N.A. Pereira, Plant natural products active against snake bite the molecular approach, *Phytochemistry* 55 (2000) 627–642, [https://doi.org/10.1016/S0031-9422\(00\)00229-6](https://doi.org/10.1016/S0031-9422(00)00229-6).